The Intramonth Momentum Cycle*

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Abstract

U.S. equity momentum returns concentrate in six trading days each month, ending four days before month-end. We trace this to predictable month-end cash demand: investors sell loser stocks for liquidity. Over 1980–2025, \$1 in a value-weighted Winners-Minus-Losers strategy grows to \$18.78 in these six days, versus \$2.37 during the rest of the month. The effect is driven by bottom-decile losers underperforming the market, while winners show no comparable concentration. The SEC's 2024 T+2-to-T+1 settlement transition shifts the selling pressure one day later, as predicted. The loser-driven pattern appears across 19 developed markets. Momentum returns largely reflect predictable month-end liquidity demand.

JEL Classification: G11, G12, G14, G23

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1 Introduction

"... Most prominent is the momentum in short-term returns, ... which in my view is the biggest challenge to market efficiency."

— Eugene F. Fama, "Two Pillars of Asset Pricing," Nobel Prize Lecture (2014)

Momentum, the tendency of past winner stocks to outperform past loser stocks, is the most pervasive anomaly in financial markets. Since Jegadeesh and Titman (1993), it has been documented in equities across dozens of markets, and momentum strategies have become a cornerstone of systematic factor investing, anchoring industry mandates that exceed USD 6 trillion globally.[1] Yet, despite three decades of research, there is no consensus on its origin. Behavioral theories trace momentum to gradual information diffusion and investor underreaction; risk-based theories argue the returns compensate for exposure to priced firm-level or macroeconomic risks.[2]

We uncover a simple financial-plumbing mechanism that accounts for the majority of momentum returns. We document that U.S. equity momentum returns concentrate in a six-day window ending four trading days before month-end. Over 1980–2025, $1 invested in the value-weighted (VW) Winners-Minus-Losers (WML) strategy during these six trading days would have grown to $18.78, whereas the same dollar invested during the rest of the month would have grown to $2.37 (Figure 1). Equivalently, 29% of trading days earn 77% of WML's cumulative return. The effect is driven by losers: bottom-decile portfolios underperform the market by 7.9 basis points per day during this six-day pre-turn-of-the-month (PreTOM) window, while winners show no comparable concentration (Figure 2).[3]

The PreTOM component of momentum returns has not weakened over time. Instead, the apparent post-2000 attenuation of U.S. momentum is driven entirely by the rest-of-month component.[4] The same loser-driven pattern appears across 19 developed markets, indicating that the mechanism is not specific to the U.S.

[1] See Gupta and Doole (2025).

[2] See, e.g., Hong and Stein (1999) and Daniel et al. (1998) for behavioral explanations; see Johnson (2002), Chordia and Shivakumar (2002), and Cochrane et al. (2008) for risk-based explanations.

[3] The PreTOM window follows Etula et al. (2020). A within-month permutation test confirms PreTOM as the deepest trough in loser market-adjusted returns ($p = 0.012$; Internet Appendix Section IA.6).

[4] Bhattacharya et al. (2017) documents the post-2000 attenuation of U.S. momentum.

Etula et al. (2020) document pre-turn-of-the-month price pressure linked to institutional investors' month-end liquidity needs. We show that this pressure falls disproportionately on momentum losers, generating the six-day concentration of WML returns. Stock-level regressions with stock and day fixed effects confirm the pattern. The PreTOM price pressure is strongest among liquid losers, consistent with liquidity-driven rather than information-driven selling. Other evidence also supports the role of month-end cash management. Trade and Quote (TAQ) data show elevated net selling pressure on losers during PreTOM, and mutual fund flow-driven selling pressure concentrates on momentum losers during the same window. Settlement-cycle changes in the U.S. and Europe shift the timing of the price pressure, linking it to investors' month-end liquidity needs.

We introduce the concept of asset dispensability, which captures how attractive a stock is to liquidate when an investor needs cash. The concept unifies three selling motives that the literature has largely treated separately. Stocks with embedded losses are attractive to sell for tax reasons; non-dividend-paying stocks are easier to liquidate because selling them gives up little expected cash income; and managers under time pressure may reach first for salient underperformers. These motives tend to point to the same stocks: momentum losers often have embedded losses, are more likely to be non-dividend payers, and are the most salient underperformers in the portfolio. A cleaner proxy for dispensability, a stock's distance from its 52-week high, produces an even larger PreTOM premium.[5] The same pattern appears in mutual fund flows: among funds with similar past returns, PreTOM outflow pressure is strongest for those farthest from their 52-week high. This evidence suggests that the poor PreTOM returns of losers operate through dispensability rather than past-return ranking itself.

To the extent that the PreTOM loser returns reflect transitory price pressure from liquidating the most dispensable stocks, rather than fundamentals, the underperformance of losers should reverse once the cash-raising window ends. Indeed, about 70 percent of the underperformance reverses within a week. TAQ order-flow data show a corresponding pattern: loser-specific selling pressure continues through the remaining month-end days and then abates in the first trading days of the next month. The remaining 30 percent persists

[5] George and Hwang (2004) use this measure and attribute its stronger and more persistent momentum returns to behavioral anchoring on the salient reference price. The paper focuses on the average cross-sectional return premium and does not examine intramonth timing.

because momentum decile membership is sticky: stocks that stay in the loser cohort accumulate repeated PreTOM pressure month after month. The residual component also unwinds, but only gradually over subsequent months and years, with some of the recovery realized after the stocks have exited the loser portfolio.

Settlement-cycle reforms provide direct evidence that settlement timing shapes the calendar concentration. The SEC's May 2024 transition from T+2 settlement, when trades settle two business days after execution, to T+1 settlement, when they settle in one business day, shifted loser-selling pressure one trading day closer to month-end. Loser returns improve four trading days before month-end, the old settlement-implied selling day, and deteriorate three trading days before month-end, the new one. This is what the mechanism predicts because shorter settlement lets investors sell one day later and still receive settled cash by month-end. The post-reform sample is short, so the individual day-level changes are imprecise, but both move in the predicted direction and their difference is statistically significant. A synchronized European T+3-to-T+2 transition provides the same test at the preceding settlement boundary, shifting loser-selling pressure from five to four trading days before month-end, with nearly all country estimates having the predicted sign. The same one-day movement appears in the Fama–French UMD factor and in past-loser mutual funds, while placebo tests on adjacent trading days and false reform dates show no comparable effect.

The dispensability mechanism extends beyond predictable monthly cycles. Dispensability operates whenever investors face cash demand, not only when it is predictable. Consistent with this, loser stocks underperformed winners by 10–18 percentage points across three dash-for-cash episodes (September 2008, October 2008, and March 2020).[6] The same asymmetry appears in mutual fund returns, most strongly in March 2020: past-loser funds returned −25% against −11% for past-winner funds, while the Fama–French UMD factor earned +9.4% the same month. After controlling for systematic factor exposure, past-loser funds underperformed their factor-implied returns by 11 percentage points in March 2020, while past-winner fund abnormal returns were essentially zero.[7]

[6]Throughout the paper we sort actively managed U.S. equity funds each month-end into deciles by trailing 12-month return; "loser funds" refers to funds in the bottom decile (D1) and "winner funds" refers to funds in the top decile (D10). Fund-level quantities are aggregated as value-weighted portfolio returns unless noted otherwise. See Section 6.1 for details.

[7]Barone et al. (2022) and Duffie (2023) study how the global dash-for-cash at the onset of COVID-19

The intramonth timing of momentum returns also refines Carhart's (1997) interpretation of mutual fund return persistence. Carhart shows that past-loser funds continue to underperform because they load negatively on momentum and charge higher expenses. The UMD factor inherits the loser-driven calendar structure that we document, earning 5.51 basis points per day during PreTOM against 1.36 during the rest of the month. Our mechanism predicts that the momentum-loading component of loser-fund underperformance should be realized mainly in that window. This is precisely what we find: past-loser fund underperformance concentrates in PreTOM and disappears once loser-stock returns are controlled for. What remains is the expense-related drag, which accrues throughout the month. The decomposition links Carhart's loser-fund persistence result to the monthly cash-demand mechanism.

Momentum crashes occur in a different part of the month: crash days concentrate at month-start, not during PreTOM, ruling out crash-avoidance risk as the source of the six-day premium. This intramonth separation complements Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015) in that the profit window and the crash window are distinct parts of the monthly cash cycle.

Our paper relates most directly to the literature on slow-moving capital (e.g., Mitchell et al., 2007; Bogousslavsky, 2016; Duffie, 2020) and to the literature that links persistent mutual fund flow pressures to momentum, such as Lou (2012) and Vayanos and Woolley (2013). In Vayanos and Woolley (2013), investors infer information about mutual funds' efficiency from their past returns, and allocate less capital to managers they view as inefficient. Momentum arises if mutual fund investors exhibit inertia.[8] Demand-system frameworks (Koijen and Yogo, 2019; Gabaix and Koijen, 2021) provide infrastructure for studying how investor flows shape asset prices; our results document a specific flow-driven mechanism behind momentum. Pitkäjärvi et al. (2020) links time-series momentum to flow-related cross-asset signals. Grinblatt et al. (1995), Sias (2004), and Puckett and Yan (2011) document momentum-chasing behavior by institutions. Lou and Polk (2022) examines how

affected Treasury and sovereign bond markets.

[8]Our findings that mutual fund flows are most negative during PreTOM on momentum-loser funds and that flow-generated selling pressure is highest during PreTOM on momentum-loser stocks are consistent with the Vayanos and Woolley (2013) information-based theory. These results are presented in the Internet Appendix. In contrast to their information channel, we find that mutual fund outflows occur mostly pre-turn-of-the-month and at the turn-of-the-month, and they are linked not only to funds' momentum but also to funds' dispensability (measured by their distance to the 52-week high) and to the dividends that the funds pay.

institutions' arbitrage in momentum strategies itself affects returns.

A second body of work documents calendar structure in returns and trading behavior. Besides Etula et al. (2020), Ogden (1990) and Lakonishok and Smidt (1988) document a turn-of-month effect in returns. More broadly, adjacent work documents monthly return seasonalities (Heston and Sadka, 2008, 2010), intraday-overnight return decompositions (Lou et al., 2019; Bogousslavsky, 2021; Barardehi et al., 2026), same-weekday continuation (Keloharju et al., 2016, 2021; Da and Zhang, 2025), and other calendar structures in specific anomalies (Birru, 2018; Cao et al., 2021; Hartzmark and Solomon, 2013, 2025). This research complements our findings, but is largely distinct from the month-end seasonalities on momentum stocks that we identify. Harvey et al. (2025) shows that investors' rebalancing between bonds and stocks around the turn-of-the-month also creates marketwide price pressures. Finally, Kutai et al. (2025) shows that settlement mechanics shape trading outcomes during stress.

Two broader literatures provide context. First, institutional frictions and the limits of arbitrage (Grossman and Miller, 1988; Shleifer and Vishny, 1997; Brunnermeier and Pedersen, 2009; He and Krishnamurthy, 2013; Adrian et al., 2014; He et al., 2017) shape asset prices and prevent arbitrageurs from eliminating the predictable return patterns that we document. The fact that momentum predictability is highest during PreTOM may reflect scarce arbitrage capital at that time, as other trading opportunities, described e.g. in Etula et al. (2020) and Harvey et al. (2025), offer high returns. Second, our work is related to the literature that documents asymmetry of anomaly returns: Stambaugh et al. (2012), Hong et al. (2000), Avramov et al. (2013), and Engelberg et al. (2018) show that anomaly profits including momentum concentrate on the short leg, attributed variously to slower information diffusion, credit distress, and news clustering.

The remainder of the paper documents the return concentration, identifies the selling-pressure mechanism, and tests its predictions across settlement reform, fund-level persistence, and momentum crashes.

2 Data and Methodology

2.1 Data Sources

We use five primary data sources. First, for both portfolio-level and stock-level analysis, we construct daily momentum decile portfolios from CRSP using fixed monthly sorts, following the Fama and French (1996) construction: all NYSE, AMEX, and NASDAQ stocks are ranked into deciles based on cumulative returns over months −12 through −2, using NYSE return breakpoints. Decile assignments are held constant within each calendar month. We obtain daily Fama-French factors and the risk-free rate from Kenneth French's data library. Dividend-paying status and distribution dates come from the CRSP distribution file.[9]

Our sample begins in 1980, consistent with the modern institutional-flow literature (Lewellen, 2011; Lou, 2012; Coval and Stafford, 2007). The stock-level panel merges the CRSP daily return panel with momentum decile assignments and bid-ask quotes, yielding 53.3 million stock-day observations over 1980–2025.

Second, for direct evidence on selling pressure, we obtain daily measures from the WRDS Intraday Indicators dataset, derived from NYSE TAQ data and covering 2003–2022. From these we construct daily net selling pressure for each stock in our sample.

Third, we use daily fund flows from Morningstar matched with quarterly fund holdings from the CRSP Mutual Fund Holdings database, spanning 2010–2023. The matched sample covers approximately 30% of total net equity assets of funds in Morningstar and CRSP combined.

Fourth, we use the CRSP Mutual Fund Database for fund-level evidence on Carhart persistence. Monthly returns span 1963–2025 (long-horizon replication); daily returns span September 1998–2025 (within-month decomposition). The sample restricts to actively managed U.S. equity diversified funds (CRSP objective code ED%, excluding index funds, ETFs, and ETNs). The monthly Carhart replication imposes no AUM threshold; the daily-NAV analysis additionally requires AUM of at least $10 million to remove very small share classes.

Fifth, for international evidence, we construct analogous daily momentum decile portfolios for 20 developed markets using Compustat Global, applying size-conditional within-country breakpoints. We report the 19 conventional momentum markets in the pooled tests

[9] Full data construction and sample formation details are in Internet Appendix Section IA.1.

and report Japan separately as a reversed-sign reference. The same Compustat Global panel supplies the fifteen European markets that migrated from T+3 to T+2 settlement in October 2014, which we use for the cross-market settlement test in Section 5.2; returns for that test are in local currency. Construction details and country-by-country results appear in Internet Appendix Section IA.9.

Although our main analysis sample begins in 1980, the cross-sectional pattern of selling losers predates it. Asem (2009) documents dividend-driven differences in momentum portfolios from 1927, and loser underperformance is well-documented across the long sample. What our analysis shows is the emergence of *intramonth timing* in this underperformance: losers underperform the market across the full month before 1962, but the underperformance increasingly concentrates in PreTOM thereafter, alongside the rise in U.S. institutional equity ownership from below 10% pre-1962 to 67% by 2010 (Blume and Keim, 2012). The PreTOM concentration first becomes statistically significant in 1962–1980 (D1 loser−Mkt difference: −5.45 bps per day, $t = -2.79$ value-weighted; −6.26 bps per day, $t = -2.90$ equal-weighted), the conventional starting point of the momentum literature (Jegadeesh and Titman, 1993). Internet Appendix Section IA.10 reports the full subperiod evidence.

2.2 Key Variables

Trading-day index and calendar windows. We index trading days relative to the turn-of-the-month (TOM). Let τ denote the last trading day of month m, and let $\tau+i$ denote the trading day i trading days after it: $\tau-1$ is the preceding trading day, $\tau-2$ the day before that, and so on, while $\tau+1$ is the first trading day of the next month. *PreTOM* is the six-day window $\tau-9, \ldots, \tau-4$, inclusive; for compactness, we also write this window as $[\tau-9, \tau-4]$. *Rest* includes all trading days outside PreTOM. *Post* denotes the seven-day window $[\tau-3, \tau+3]$ around month-end. Throughout the paper, we use the terms "turn-of-the-month" and "month-end" interchangeably.

Momentum portfolios. We define WML as the daily value-weighted (VW) return on the top momentum decile (D10) minus the daily value-weighted return on the bottom decile (D1), where value weights within each (day, decile) cell are previous-day (one-day lagged) market capitalizations normalized to sum to one. We also report equal-weighted (EW) variants. We write $D_{i,t} \in \{1, \ldots, 10\}$ for the momentum decile assignment of stock i on day t; assignments are formed monthly and held constant within each calendar month.

Bid-ask spread. From CRSP daily bid and ask quotes, we compute the daily percentage bid-ask spread as $(\text{Ask}_{i,t} - \text{Bid}_{i,t})/\text{Midpoint}_{i,t}$. We aggregate this to a stock-month measure $\text{BAS}_{i,m-1}$ defined as the average daily spread of stock i over the prior calendar month $m-1$; stock-months with fewer than 10 valid daily BAS observations in month $m-1$ are dropped. We use the pre-period (lagged) measure rather than the contemporaneous spread so that the regressor is predetermined relative to day-t returns, avoiding any mechanical correlation between intraday spread movements and same-day returns.

Selling pressure. From the WRDS Intraday Indicators dataset described above, we construct daily net selling pressure (sell share minus buy share of dollar volume).

Settlement transitions. U.S. equity settlement shortened from T+3 to T+2 on September 5, 2017, and from T+2 to T+1 on May 28, 2024.[10] In Europe, fifteen markets in our sample shortened settlement from T+3 to T+2 on October 6, 2014, under the EU Central Securities Depositories Regulation. We use these transitions for the difference-in-differences (DiD) analysis of Section 5.

Notation. Throughout the paper, lowercase r denotes stock, market, or factor returns and capital R denotes fund returns; r_t^m is the market return and r_m^f the risk-free rate. Subscripts identify stock i, fund j, country c, trading day t, and month m.

3 Main Results

3.1 Portfolio-Level Evidence

We begin with the value-weighted WML portfolio and decompose its cumulative return by trading-day position within the month. We classify each trading day as PreTOM or Rest and

[10]Both transitions were mandated by amendments to SEC Rule 15c6-1(a): the T+3 to T+2 shift in 2017 (Securities and Exchange Commission, 2017) and the T+2 to T+1 shift in 2024 (Securities and Exchange Commission, 2023).

compound returns within each window separately from \$1 on January 2, 1980.[11] Figure 1 shows a stark divergence: by 2025, the PreTOM-only strategy has grown to \$18.78 while the Rest-only strategy reaches just \$2.37.[12]

The asymmetry in returns is overwhelmingly loser-driven. Figure 2 plots average daily market-adjusted returns for momentum winners, losers, and the WML spread during PreTOM and Rest. Losers underperform the market by 7.9 basis points per day during PreTOM ($t = -3.72$) but earn returns an order of magnitude smaller and indistinguishable from zero during Rest (-0.6 bps per day, $t = -0.43$). Winners earn small, statistically insignificant returns in both windows and show no comparable PreTOM differential. Consequently, the WML spread itself is insignificant during Rest ($+2.5$ bps per day, $t = 1.33$). Momentum premiums are mostly about losers losing at predictable times and less about winners winning. This asymmetry is robust across subperiods.[13]

Table 1 formalizes the portfolio-level results in Figure 2 using daily Fama–MacBeth cross-sectional stock-level regressions. For each trading day t, we regress market-adjusted stock returns on momentum-decile indicators:

$$r_{i,t} - r_t^m = \sum_{d=1}^{10} b_{d,t} \, \mathbf{1}\{D_{i,t} = d\} + u_{i,t}, \tag{1}$$

where $r_{i,t}$ is stock i's daily return on day t, r_t^m is the value-weighted CRSP market return that day, $D_{i,t}$ is stock i's momentum decile on day t, and $\mathbf{1}\{D_{i,t} = d\}$ is an indicator equal to one if stock i is in decile d on day t. The regression has no intercept and weights observations by previous-day market capitalization, so $b_{d,t}$ is the value-weighted market-adjusted return of momentum decile d on day t.

We then test whether the loser and winner decile returns differ across calendar windows

[11] WML is a self-financing long-short portfolio; on days outside the relevant window the strategy holds cash at zero return.

[12] Using French's daily-rebalanced momentum portfolios yields \$15.52 and \$2.07 respectively. Daily rebalancing reassigns deciles every day on a sliding 12-month formation window, so the cohort defined as "losers" shifts continuously. Our fixed monthly sort holds the cohort constant within each calendar month, which more closely tracks the institutional positions that face month-end selling pressure. The qualitative pattern is identical under either convention.

[13] Internet Appendix Figure IA.12 plots the same decomposition for the 1980–2002 and 2002–2025 halves of the sample; both show the loser PreTOM underperformance and WML PreTOM concentration.

by regressing the daily series of $b_{d,t}$ on a PreTOM indicator:

$$b_{d,t} = c_0^d + c_1^d \,\mathrm{PreTOM}_t + v_{d,t}, \qquad d \in \{1, 10\}, \tag{2}$$

where c_0^d is the average market-adjusted return of decile d (1 or 10) on Rest days; $c_0^d + c_1^d$ is the PreTOM average; c_1^d is the PreTOM–Rest differential. Table 1 reports the estimates. For losers ($d = 1$), the Rest-day average is $c_0^1 = -0.87$ basis points per day and the PreTOM average is $c_0^1 + c_1^1 = -7.98$ basis points; the PreTOM–Rest differential is $c_1^1 = -7.12$ ($t = -2.79$). For winners ($d = 10$), the Rest-day average is +1.52 bps, the PreTOM average is +2.17 bps, and the differential is $c_1^{10} = +0.65$ ($t = 0.39$). The PreTOM concentration is one-sided, overwhelmingly on the loser leg, which is difficult to reconcile with explanations that act symmetrically on winners and losers, including symmetric risk-based or mispricing accounts.

The remaining analyses use the stock-level panel framework, which accommodates stock and day fixed effects, multi-way interactions, and stock-day covariates.

3.2 Stock-Level Evidence

3.2.1 PreTOM Loser Underperformance

The portfolio evidence shows the concentration is loser-driven. We now test this at the stock level on the full panel of 53.3 million CRSP common-stock-day observations (1980–2025), where stock and day fixed effects absorb confounds that portfolio aggregation cannot. We estimate the panel regression

$$r_{i,t} - r_t^m = \beta_1 \,\mathrm{Loser}_{i,t} + \beta_2 \,\mathrm{Loser}_{i,t} \times \mathrm{PreTOM}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \tag{3}$$

where $r_{i,t} - r_t^m$ is the daily market-adjusted return of stock i on day t (in basis points), with $r_{i,t}$ being the stock's daily return and r_t^m being the value-weighted CRSP market return that day. $\mathrm{Loser}_{i,t}$ is an indicator equal to one if stock i is in the bottom momentum decile on day t. PreTOM_t is an indicator equal to one if day t falls in the $[\tau-9, \tau-4]$ window; the standalone PreTOM indicator is absorbed by δ_t. μ_i and δ_t are stock and day fixed effects, and standard errors are double-clustered by stock and day. The coefficient of interest is β_2, which represents the additional underperformance of losers on PreTOM days relative to losers outside PreTOM. We also report equal-weighted (EW) estimates.

Table 2 presents the baseline estimates. In the value-weighted specification, the Loser × PreTOM coefficient is $\beta_2 = -7.15$ basis points per day ($t = -3.08$). This confirms that loser returns are significantly lower during PreTOM. The equal-weighted β_2 is smaller (-2.55 bps, $t = -1.85$). The gap between value-weighted and equal-weighted estimates previews the bid-ask-spread evidence below: the PreTOM effect concentrates where institutions trade, and equal-weighting dilutes it across illiquid micro-caps where they do not.

3.2.2 Liquid Losers Drive the PreTOM Underperformance

We now ask whether the PreTOM hit on losers concentrates among liquid D1 stocks, the prediction implied by an institutional-selling channel. We restrict the sample to bottom-decile stocks and estimate

$$r_{i,t} - r_t^m = \beta\,\mathrm{PreTOM}_t + \gamma_1\,\mathrm{BAS}_{i,m-1} + \gamma_2\,\mathrm{PreTOM}_t \times \mathrm{BAS}_{i,m-1} + \mu_i + \delta_{m(t)} + \varepsilon_{i,t}, \quad i \in \mathrm{D1}, \tag{4}$$

where $r_{i,t} - r_t^m$ is stock i's daily market-adjusted return (in excess of the value-weighted CRSP market return, as in equation (3)), and $\mathrm{BAS}_{i,m-1}$ is the average daily bid-ask spread of stock i over the prior calendar month $m - 1$ (predetermined relative to day-t returns). μ_i are stock fixed effects, and $\delta_{m(t)}$ are year-month fixed effects, allowing the standalone PreTOM coefficient to be identified within month. Standard errors are double-clustered by stock and day. The coefficient γ_2 measures how the within-D1 PreTOM effect varies with the ex-ante bid-ask spread. γ_2 should be positive if PreTOM selling concentrates in the most liquid losers, those with the smallest bid-ask spread.

Table 3 reports the estimates with the bid-ask spread measured ex ante as the stock's average BAS over the prior calendar month. The interaction γ_2 is positive in both specifications: $+39.04$ ($t = 1.41$) in value weighting (column 1) and $+29.54$ ($t = 2.82$, significant at the 1% level) in equal weighting (column 2). The point estimates imply that the PreTOM loser hit attenuates as the bid-ask spread rises, consistent with institutions preferentially executing PreTOM selling in low-spread (liquid) losers where transaction costs are lowest. The equal-weighted specification identifies the interaction most cleanly, consistent with small stocks contributing substantial cross-sectional variation in bid-ask spreads; under value weighting, the spread distribution is compressed toward zero and the interaction is identified less precisely. Under both weighting schemes, the qualitative pattern supports the dispensability interpretation of the cross-sectional mechanism we identify.

3.2.3 The PreTOM Effect Persists and Strengthens Over Time

A natural concern is that the PreTOM effect reflects an artifact of earlier, less liquid markets, such as thin trading, wider spreads, and less institutional participation, that has since disappeared as markets modernized. Many well-documented anomalies do attenuate over time as arbitrage capital grows and markets become more efficient (McLean and Pontiff, 2016; Greenwood and Sammon, 2025). If so, the PreTOM effect should be stronger in the first half of the sample and attenuate or vanish in the second. Yet the opposite is true.

Table 4 re-estimates equation (3) on the two halves of the sample, split at the midpoint (July 2002). The Loser×PreTOM coefficient appears in both halves and strengthens over time: $\beta_2 = -5.76$ bps ($t = -2.31$) in 1980–2002 and $\beta_2 = -8.50$ bps ($t = -2.20$) in 2002–2025.

4 The Mechanism

4.1 Selling Pressure During PreTOM

Section 3.2.2 showed that the PreTOM effect concentrates in liquid losers. We now use intraday trade-direction data to test directly whether losers experience elevated selling pressure during PreTOM. We construct the data from the WRDS TAQ Intraday Indicators dataset, which provides daily measures of buy- and sell-initiated volume classified by the Lee-Ready algorithm. The sample covers 2003–2022, yielding approximately 17.9 million stock-day observations.

We estimate the analogue of equation (3) with selling pressure as the dependent variable:

$$\text{NetSell}_{i,t} = \beta_1 \,\text{Loser}_{i,t} + \beta_2 \,\text{Loser}_{i,t} \times \text{PreTOM}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \tag{5}$$

where $\text{NetSell}_{i,t}$ is daily net selling pressure (sell volume minus buy volume, divided by total volume), which by construction ranges from -1 (all volume buyer-initiated) to $+1$ (all volume seller-initiated); higher values indicate more selling pressure. μ_i and δ_t are stock and day fixed effects, and standard errors are double-clustered by stock and day. β_2 captures how much more strongly losers are sold on PreTOM days than on Rest days.

Table 5 reports estimates of equation (5). On PreTOM days, losers experience an additional 0.2 percentage points of net selling pressure relative to non-losers, beyond the Rest-day baseline gap ($\beta_2 = +0.0020$, $t = 2.50$).[14]

A complementary measure of selling pressure comes from mutual fund flows. Following Coval and Stafford (2007) and Lou (2012), we construct stock-specific flow pressures by attributing each fund's daily flows to its holdings as of the previous quarter-end, then aggregating across funds and normalizing by market capitalization. Figure 3 shows average flow pressure during PreTOM by momentum decile. The pattern is monotonic: losers (D1) face the strongest outflow pressure (-9.9 basis points), while winners (D10) experience slight inflow pressure (+0.5 bps).

4.2 Why Losers Face Selling Pressure During PreTOM

Section 4.1 established that PreTOM selling pressure falls disproportionately on losers. Why losers specifically? We argue that loser stocks are often "dispensable" stocks that investors are most likely to liquidate when cash is needed. Three important channels make losers dispensable: tax-loss harvesting incentives, lower dividend income, and institutional selling behavior under time pressure.[15]

4.2.1 Tax-Related Selling

Tax considerations can make losers attractive to sell (Constantinides, 1984; Grinblatt and Moskowitz, 2004), but we do not find evidence that they fully account for the monthly PreTOM concentration. The effect remains in the months when tax and window-dressing motives should be weakest: excluding quarter-end months yields a Loser $\times$ PreTOM coefficient of -7.5 bps per day ($t = -2.66$), essentially identical to the full-sample estimate.

[14] Lee–Ready trade signing is measured with error, so signed order-imbalance estimates are attenuated toward zero. The classic benchmark is about 85% accuracy (Odders-White, 2000); recent TAQ evidence finds similar raw Lee–Ready accuracy, rising from 86% to 92% after latency adjustment (Holden et al., 2023). We therefore interpret the imbalance coefficient as conservative.

[15] The types of stocks that are most dispensable are likely to evolve over time. There is also likely endogeneity in this relation as the characteristics that make stocks dispensable generate selling pressure, depress prices and cause these stocks to be overrepresented in the momentum loser portfolio.

December, the month with the strongest tax incentives, also does not drive the PreTOM effect on its own. D1 losers do underperform during December PreTOM (−11.08 bps per day, $t = -1.84$), but removing December from the sample lowers the full-sample Loser × PreTOM coefficient only from −7.15 to −7.03 bps per day ($t = -2.86$; Internet Appendix Table IA.8, column 2).[16] While tax considerations likely play a role, the monthly PreTOM cycle is broader than any tax-driven pattern alone can explain. Full estimates are given in Internet Appendix Section IA.11.

4.2.2 Dividend Income

Losers generate less dividend income. Among D1 stock-months, 84.1% are non-payers, compared with 67.5% among D10 stock-months. Selling losers therefore entails less foregone dividend income when investors need cash. Consistent with this channel, Internet Appendix Figure IA.5, Panel A, shows that mutual-fund-flow pressure during PreTOM is more negative for non-dividend payers than for dividend payers.

Among quarterly dividend payers, we examine whether selling pressure varies within the dividend payment cycle.[17] An investor covers end-of-month obligations from two sources: incoming cash dividends and proceeds from selling stocks. The two are substitutes. We split each stock's months by their position in the payment cycle: Month 1 is the payment month, Month 2 is one month later, and Month 3 is the month before the next payment. We treat the within-quarter pattern as descriptive evidence rather than a mechanism-derived prediction, since stock-specific dividend timing need not translate into stock-specific selling pressure when institutions manage cash at the portfolio level.

The TAQ data show a monotonic pattern across the cycle. Loser net selling pressure during PreTOM is 0.18 percentage points in Month 1, rising to 0.51 in Month 2, and 0.57 in Month 3, with Month 3 statistically significant ($t = 2.12$). Winner net selling pressure during PreTOM is statistically flat across the same months ($F = 0.38$, $p = 0.68$); the depletion pattern is loser-specific. Return-based specifications and full estimates appear in

[16] A separate mid-December pattern of negative D1 returns (−13.38 bps per day in the second week, $t = -1.79$) is consistent with tax-loss harvesting timed to mutual-fund capital-gains distributions (Sialm and Starks, 2012), but operates outside PreTOM and does not eliminate the PreTOM cycle.

[17] A simpler test would compare PreTOM selling on dividend-paying stocks against non-paying stocks. 84% of D1 stock-months are non-payers versus 68% of D10 (Table IA.9, Panel A), so this comparison largely reproduces the winner-loser sort. We use within-loser-decile variation to control for selection into deciles.

Internet Appendix Section IA.12 and Table IA.9.

4.2.3 Institutional Selling of Losers

Two pieces of evidence point to institutional selling: as documented in Section 3.2.2, the PreTOM effect concentrates in liquid losers. In addition, the Internet Appendix shows that the PreTOM loser effect remains present among S&P 500 stocks, where institutions commonly hold positions (Internet Appendix Table IA.23). The broader institutional-trading literature is consistent. Unlike retail investors, who tend to hold losers (Odean, 1998), large institutions actively reduce risk following losses (O'Connell and Teo, 2009). Akepanidtaworn et al. (2023) documents that under time pressure, managers sell heuristically, reaching for the most salient positions rather than optimizing.

4.2.4 Sorting by Distance from the 52-Week High

Tax-loss harvesting, the lower dividend yields of losers, and institutional selling heuristics under time pressure point to a common feature: losers are dispensable holdings that investors are likely to liquidate first when facing month-end cash needs. If our mechanism operates on dispensability rather than on the specific construction of the standard momentum 12-2 sort (decile portfolios formed on cumulative returns over the previous 12 months), alternative loser sorts that capture dispensability should reveal the same PreTOM pattern.

One natural candidate is the 52-week high. George and Hwang (2004) documents that distance from the 52-week high produces stronger and more persistent momentum returns than the canonical 12-2 measure, and interprets this through a behavioral anchoring mechanism. Our framework offers a complementary reading: 52-week-high distance aligns with the dispensability channels above. It measures embedded losses more directly than the 12-2 sort does (the tax channel),[18] reflects recent underperformance (the institutional channel and the Grinblatt and Han disposition effect), and correlates with non-payer status more strongly than the 12-2 sort (the dividend channel).[19] The 12-2 sort captures the institutional and

[18] A stock can have a negative 12-2 return but still trade above its purchase price if it rallied during the most recent month; conversely, a stock near its 52-week low is almost certainly in unrealized-loss territory for any investor who entered during the past year. 52-week-high distance is therefore a more direct proxy for the embedded losses that drive tax-motivated selling.

[19] Sorting stocks by 52-week-high distance produces a clean monotonic pattern in the share of non-paying firms from 91.0% non-payers in the bottom decile, falling steadily to 35.4% in the top decile, a 56-percentage-

behavioral channels through past returns but only indirectly captures the tax channel. The 52-week-high sort therefore offers a more direct test of the dispensability mechanism along these dimensions.

Consistent with the 52-week-high sort capturing dispensability differences better than the traditional 12-2 momentum sort, we find that during our 45-year sample period, \$1 invested during the PreTOM periods in a long-short portfolio sorted by distance from the 52-week-high grows to \$45.72. This contrasts with the \$1 invested during PreTOM in the momentum WML portfolio that grows to \$18.78.

Both characteristics derive the bulk of their long-short returns from the same six-day PreTOM window, with little cumulative growth during the rest of the month. The similarity in timing suggests that both sorts load on the same underlying selling-pressure mechanism. The 52-week-high sort nevertheless produces larger cumulative returns, consistent with the dividend evidence that 52-week-high distance separates dispensable from non-dispensable stocks more cleanly.

The flow-pressure evidence points in the same direction. Internet Appendix Figure IA.5 shows that, during PreTOM, mutual-fund-flow pressure is more negative for non-dividend payers than for dividend payers. The pattern appears both when stocks are grouped by distance from the 52-week high and when they are grouped by fund momentum. This reinforces the interpretation that the selling pressure is not tied mechanically to the 12-2 momentum ranking itself, but to characteristics that make some losing positions more natural to liquidate when investors need cash. A double sort makes the point directly. Within loser-momentum funds, PreTOM flow pressure ranges from -9.7 for funds farthest from their 52-week high to -2.1 for funds near it (Internet Appendix Table IA.10). The momentum ranking alone does not determine which positions face the outflows.

4.3 Short-Term and Long-Term Reversal

If the PreTOM concentration reflects transitory selling pressure rather than fundamental news, the gap that opens between losers and non-losers during PreTOM should partially close once the cash-raising window ends. We extend equation (3) by adding a Loser $\times$ Post

point spread. The same sort by 12-2 returns produces a U-shape: 84.1% non-payers among losers, dropping to a trough near the seventh decile, rising again to 67.5% among winners. For additional evidence, see Figures IA.6 and IA.7 in the Internet Appendix.

interaction and regress daily market-adjusted stock returns on the loser dummy interacted with PreTOM and Post indicators, with stock and day fixed effects. The month splits into three non-overlapping windows: PreTOM $[\tau{-}9, \tau{-}4]$, Post $[\tau{-}3, \tau{+}3]$ (surrounding month-end), and mid-month days that are neither PreTOM nor Post (the omitted reference category):

$$r_{i,t} - r_t^m = \beta_1 \, \text{Loser}_{i,t} + \beta_2 \, \text{Loser}_{i,t} \times \text{PreTOM}_t + \beta_3 \, \text{Loser}_{i,t} \times \text{Post}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \quad (6)$$

where β_1 is the loser-vs-non-loser return gap on mid-month days. β_2 picks up the additional underperformance of losers on PreTOM days, i.e., the widening of the gap relative to mid-month, consistent with cash-raising selling pressure. β_3 picks up the reversal on Post days, as the pressure subsides. Day fixed effects absorb market-wide PreTOM and Post effects, so the interactions identify cross-sectional loser-specific variation. In a typical 21-day month, PreTOM accounts for 6 trading days, Post accounts for 7 trading days, and the mid-month reference window accounts for 8 trading days. Table 6 reports the estimates. The PreTOM gap widens by $\beta_2 = -5.59$ basis points per day ($t = -2.07$); the Post gap narrows by $\beta_3 = +3.36$ basis points per day ($t = 1.24$), positive but not significant. However, the difference between the two windows is large and highly significant: $\beta_3 - \beta_2 = +8.94$ basis points per day ($t = 3.30$, $p < 0.001$). Cumulatively, the Post-window narrowing of the gap ($7\beta_3 = +23.5$ basis points) recovers about 70 percent of the PreTOM widening ($6\beta_2 = -33.5$ basis points).[20] The TAQ evidence supports this abatement, but only in the later part of the Post window. Loser-specific selling pressure remains elevated through the final days of the month and unwinds only in the first days of the next month (untabulated).

The Post-window rebound on D1 stocks traces the immediate unwind of some of the price pressures from institutions' PreTOM selling that focuses on D1 stocks. Stocks that fall in D1 across multiple past PreTOM windows accumulate price pressure from multiple cycles of PreTOM selling. If most, or all, of that price pressure is transitory, the long-term accumulation of price pressure should lead to mean-reversion at longer horizons. By contrast, cumulative returns from non-PreTOM days are likely to reflect information rather than flow

[20]The main specification reassigns deciles monthly, so the same stock can leave D1 between PreTOM and the subsequent Post window. To rule out that the reversal reflects cohort turnover, we re-run the analysis tracking each month's D1 stocks through the following Post window even if they exit D1; the difference is essentially unchanged ($\beta_3 - \beta_2 = +8.74$ matched-cohort vs. $+8.94$ contemporaneous, untabulated).

and should not mean-revert.

We confirm both predictions in Internet Appendix Table IA.24: in a within-stock panel spanning all deciles (43.5 million stock-days, stock and day fixed effects),[21] past PreTOM returns accumulated while a stock was in D1 revert about three times more strongly than past PreTOM returns accumulated outside D1 (Panel B; Wald $p = 0.0006$); past non-PreTOM returns show at most weak reversion, with no asymmetry by D1 status. Panel B leaves open whether the reversion of past PreTOM-while-D1 returns is mechanically tied to current D1 membership: stocks that cycle in and out of D1 could be re-experiencing fresh PreTOM pressure each month that unwinds within the cycle. Panel C addresses this by splitting the panel on whether the stock is currently in D1. The reversion of past PreTOM-while-loser returns continues to materialize among stocks that have exited D1 ($t = -3.61$), so the imprint of past PreTOM-while-loser windows is a stock-level feature that keeps unwinding regardless of current cohort membership.

These long-horizon results refine the long-horizon reversal noted by Jegadeesh and Titman (1993), who show that momentum profits at the 3–12 month horizon gradually reverse over subsequent years. Our decomposition isolates which past returns drive that reversal: it is the PreTOM returns accumulated while a stock was in D1, consistent with the slow unwinding of transitory selling pressure.

4.4 Alternative Explanations

This section rules out three alternative explanations and outlines additional robustness. Quarter-end window dressing (Brown, 2017) is not a driver of PreTOM return concentration: restricting to non-quarter-end months produces a Loser × PreTOM coefficient of −7.5 bps ($t = -2.66$), essentially identical to the full-sample estimate.

Benchmark-mediated rebalancing is also inconsistent with both the direction and the timing of PreTOM loser underperformance. Equity funds benchmarked to a value-weighted index face no rebalancing demand after a stock's price decline: the stock's benchmark weight falls in lockstep with its portfolio weight, leaving the fund aligned with the index without

[21] The sample is smaller than the 53.3 million stock-days in our baseline panel because the long-horizon predictors require a filled rolling window of past returns from $[t-480, t-60]$; observations without sufficient history (early-sample burn-in, coverage gaps, short-lived stocks) drop out.

any trade.[22] Fixed-weight rebalancing would generate trades, but in the wrong direction: a price decline pushes a stock below its target dollar weight, prescribing a *purchase* rather than a sale. Where institutional rebalancing has been documented to generate predictable price patterns, it operates at the asset-class level (stocks vs. bonds) and concentrates in the last five trading days of the month (Harvey et al., 2025); neither fits the cross-sectional, mid-month structure of PreTOM, which is the six-day window immediately preceding their rebalancing window. The T+1 settlement reform documented in Section 5 provides causal confirmation: the underperformance window shifts with settlement-cycle rules that do not affect institutional rebalancing schedules.

Passive beta exposure is also inconsistent with the pattern. Etula et al. (2020) document a marketwide PreTOM dip, so high-beta stocks should mechanically underperform more during PreTOM. If the loser-PreTOM concentration were merely losers loading more heavily on this aggregate dip, then the Loser × PreTOM coefficient should weaken once we add a market-beta interaction. We test this directly in Internet Appendix Tables IA.13, IA.14, and IA.15, which augment the baseline VW, EW, and TAQ specifications with stock-level CAPM beta and the triple interaction Loser × PreTOM × β. Two pieces of evidence make the beta channel implausible. First, the long-short WML portfolio has essentially zero net market beta ($\beta_{WML} = -0.08$), so the marketwide PreTOM dip largely cancels in WML and cannot generate the long-short asymmetry. Second, in the saturated specification, the Loser × PreTOM coefficient is essentially unchanged from the baseline (−7.02 VW and −2.33 EW, compared with −7.15 and −2.55 in Table 2), and the triple interaction Loser × PreTOM × β is statistically indistinguishable from zero in each of the three saturated specifications (one per table).

Transaction costs do not overturn the intramonth pattern. Following Novy-Marx and Velikov (2016), we adjust returns using round-trip costs based on quoted bid-ask spreads. Over 1980–2025, a monthly rebalanced WML strategy earns 91.3 bps per month gross and 11.5 bps net; 65% of the gross premium accrues during PreTOM. After decimalization, when quoted spreads narrow substantially, the concentration strengthens: over 2001–2025, a PreTOM-only decomposition earns 54.1 bps per month gross and 14.9 bps net, whereas

[22] Index deletions do generate forced selling by passive funds, but they operate on quarterly or annual reconstitution schedules rather than the monthly cycle we document, and apply to a small fraction of D1 stocks in any given month. Consistent with this, the PreTOM loser effect is, if anything, *stronger* among S&P 500 constituents (Internet Appendix Table IA.23), where deletion risk is negligible.

the Rest-only decomposition earns −66.1 bps net. Thus, in modern markets, the net WML premium is concentrated in the PreTOM window. The result describes when the momentum premium is realized, not a within-month trading rule. Internet Appendix Section IA.3 reports the full decomposition.

Additional robustness checks, including S&P 500 membership splits and alternative holding periods, are reported in the Internet Appendix.

5 Settlement-Cycle Evidence

The preceding sections show that loser underperformance concentrates before month-end and is accompanied by loser-specific selling pressure. This section tests a more specific implication of the cash-management mechanism. If investors sell loser stocks to raise settled cash for month-end obligations, then the timing of the selling window should depend on the equity settlement cycle. A one-day shortening of the settlement cycle should move the marginal cash-raising day one trading day closer to month-end.

The prediction is mechanical. Under a longer settlement cycle, investors who need settled cash by month-end must sell earlier. Under a shorter settlement cycle, they can sell one day later and still receive cash by the same deadline. A transition from T+3 to T+2 should therefore move the marginal cash-raising day from $\tau-5$ to $\tau-4$, while a transition from T+2 to T+1 should move it from $\tau-4$ to $\tau-3$.[23] The test is the relative movement of adjacent boundary days, not the unconditional level of returns on any single day.

We test this prediction using three settlement-cycle reforms. The May 2024 U.S. transition from T+2 to T+1 provides the main within-market test. A synchronized European transition from T+3 to T+2 on October 6, 2014 provides a cross-market replication of the earlier settlement shortening. The September 2017 U.S. transition from T+3 to T+2 provides an additional directional check. Across these settings, the identifying comparison is the same: when settlement shortens, the marginal selling pressure relocates from the old boundary day to the new one, so after the reform loser returns should rise on the old boundary day and fall on the new boundary day.

[23]The boundaries above are relative to the unconditional window $[\tau-9, \tau-4]$, estimated over the full sample. This trough is stable: it stays at $[\tau-9, \tau-4]$ when the post-T+1 period is excluded. The six-day window is a smoothed aggregate that does not track one-day shifts in the marginal selling day; those shifts are identified by the reform difference-in-differences below.

5.1 U.S. T+2-to-T+1 Transition

We begin with the SEC's May 28, 2024 transition from T+2 to T+1 equity settlement. Under the cash-management mechanism, shortening settlement by one day should shift the marginal cash-raising day from $\tau-4$ to $\tau-3$. Before the reform, investors who needed settled cash by month-end had to complete the marginal sale by $\tau-4$. After the reform, they could complete the same sale at $\tau-3$ and still receive settled cash by the relevant deadline.

Figure 4 plots the post-vs-pre difference in mean loser market-adjusted returns at each trading day relative to month-end. The pattern is concentrated at the predicted boundary days. At $\tau-4$, loser returns improve after the reform; at $\tau-3$, loser returns deteriorate. Other nearby days show no comparable movement. This is the one-day relocation predicted by the settlement mechanism.

The DiD compares $\tau-4$ and $\tau-3$ against early-month controls $[\tau+5, \tau+8]$, which experience the same post-2024 time-series shocks but are plausibly unaffected by settlement mechanics. Days closer to month-end $[\tau-2, \tau-1]$ are contaminated by post-PreTOM reversal and thus unsuitable as controls.[24] We estimate the following portfolio-level time-series regression on the D1 loser portfolio:

$$\begin{aligned} R_t^{D1} - r_t^m = \alpha + \beta_{\tau-4}\,\mathbf{1}_{[\tau-4]} + \beta_{\tau-3}\,\mathbf{1}_{[\tau-3]} + \pi\,\mathrm{T1Reform}_t \\ + \theta_{\tau-4}\,\mathbf{1}_{[\tau-4]} \times \mathrm{T1Reform}_t + \theta_{\tau-3}\,\mathbf{1}_{[\tau-3]} \times \mathrm{T1Reform}_t + \varepsilon_t, \end{aligned} \tag{7}$$

where $R_t^{D1} - r_t^m$ is the daily value-weighted market-adjusted return on the D1 loser portfolio on date t (one observation per trading day), $\mathrm{T1Reform}_t$ indicates dates from June 1, 2024 onward, and the omitted reference is $[\tau+5, \tau+8]$. The sample collapses to three observations per month ($\tau-4$, $\tau-3$, and the early-month $[\tau+5, \tau+8]$ control mean) and is estimated with Newey-West standard errors. May 2024 is excluded from both pre- and post-samples to avoid contamination at the regime boundary. The prediction is $\theta_{\tau-4} - \theta_{\tau-3} > 0$.

Table 7, Panel A, reports raw cell means: the loser mean at $\tau-4$ shifts from -6.9 bps per day pre-reform to $+51.4$ post-reform; at $\tau-3$, from -3.7 to -31.4; the early-month control

[24] Using $\tau-2$ as the control yields similar estimates (untabulated). The right edge of the pre-reform cash-raising window has been independently identified at $\tau-4$: see Etula et al. (2020) and Internet Appendix Figure IA.2, which shows that $[\tau-9, \tau-4]$ is the unique trough in loser market-adjusted returns across the second half of the trading month in the full 1980–2025 sample. The $\tau-4$ to $\tau-3$ prediction is thus not a consequence of inherited window definitions.

shifts only modestly. Panel B converts these into DiD coefficients. The two days move in opposite directions, neither individually significant given the 19-month post-reform sample. Pooling them tightens the test: $\theta_{\tau-4} - \theta_{\tau-3} = +85.9$ basis points ($t = 2.68$, $p = 0.007$).[25] Restricting the pre-period to the T+2 era alone (September 2017 to April 2024) yields a similar result with +85.7 ($t = 2.31$).

Panel B isolates the time-of-month shift but cannot rule out a generic post-2024 repricing affecting all stocks at $\tau-4$ and $\tau-3$. Panel C addresses this with a stock-level triple-difference contrasting D1 against D10 at the same days:

$$
\begin{aligned}
r_{i,t} - r_t^m = {} & \theta_{\tau-4}\,\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-4]} \times \text{T1Reform}_t + \theta_{\tau-3}\,\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-3]} \times \text{T1Reform}_t \\
& + Z'_{i,t}\gamma + \mu_i + \delta_t + \varepsilon_{i,t},
\end{aligned} \tag{8}
$$

where $Z_{i,t}$ contains the lower-order $\text{Loser}_{i,t}$, $\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-m]}$, and $\text{Loser}_{i,t} \times \text{T1Reform}_t$ terms; the $\mathbf{1}_{[\tau-m]}$, T1Reform_t, and $\mathbf{1}_{[\tau-m]} \times \text{T1Reform}_t$ effects are absorbed by δ_t. The sample is restricted to days $\tau-4$, $\tau-3$, and the early-month control window, with early-month days as the omitted day reference and D10 as the omitted decile reference.

Table 7, Panel C, reports the estimates. The shift is loser-specific: $\theta_{\tau-4} - \theta_{\tau-3} = +196$ basis points ($t = 2.20$, $p = 0.028$). The magnitude exceeds the loser-only estimate in Panel B (+85.9 bps) because the triple-difference captures opposing movement at the winner leg. Under the dash-for-cash mechanism, funds sell losers preferentially to raise cash while retaining winners, so the loser–winner spread widens by approximately 200 basis points across the two boundary days. The result rules out a generic post-2024 repricing common to all stocks at $\tau-4$ and $\tau-3$ and identifies the boundary-day shift as a redistribution of trading pressure between the two legs of the momentum sort, consistent with fund cash-management around month-end.[26]

[25] We also test for parallel pre-trends in an event study that replaces the T1Reform_t dummy variable with twelve-month event-time bins, eight before the reform and two after, with the year just before the reform as the omitted reference. The eight pre-reform coefficients are jointly indistinguishable from zero ($F(8, 1080) = 0.59$, $p = 0.78$), and both post-reform coefficients are close to the pooled estimate (+121 bps, $t = 2.00$; +111 bps, $t = 1.81$). See Internet Appendix Figure IA.1.

[26] Placebo tests on non-boundary days ($\tau-6$ vs. $\tau-7$) and on placebo regime dates (May 28 of 1999, 2005, 2011, 2013, and 2015) show no effect (Internet Appendix Table IA.3).

5.2 Synchronized European T+3-to-T+2 Transition

On October 6, 2014, fifteen European markets in our sample migrated from T+3 to T+2 settlement under the EU Central Securities Depositories Regulation (CSDR) (European Central Securities Depositories Association, 2014; International Capital Market Association, 2014). The last T+3 trade date was Friday, October 3, 2014, and from Monday, October 6, 2014, settlement was based on T+2 (European Central Bank, 2014). This synchronized transition provides a cross-market test of the settlement-clock mechanism.

The starting point is that European loser returns display the same broad PreTOM trough as U.S. loser returns: pooled across these fifteen markets, loser excess returns are lowest over $[\tau{-}9, \tau{-}4]$, both over the full sample and in the pre-reform T+3 period (Internet Appendix Figure IA.9). This unconditional window identifies where average selling pressure is concentrated. The settlement test is more local. It asks whether the reform reallocates selling pressure between the adjacent days implied by the old and new settlement regimes. For a T+3-to-T+2 transition, that comparison is $\tau{-}5$ versus $\tau{-}4$. Under T+3, investors facing the relevant month-end cash deadline must complete the marginal sale one day earlier than under T+2. The mechanism therefore predicts that, after the reform, loser returns should improve on $\tau{-}5$, the old marginal cash-raising day, and deteriorate on $\tau{-}4$, the new marginal cash-raising day.

We estimate the boundary-day contrast separately by country and aggregate the country-level estimates using inverse-variance weights. This treats the country as the replication unit and avoids relying on panel standard errors with a small number of country clusters.

For each country c, we estimate

$$\begin{aligned} R^{D1}_{c,t} - r^{m}_{c,t} = \alpha_c + \beta_{c,\tau-5}\,\mathbf{1}_{[\tau-5]} + \beta_{c,\tau-4}\,\mathbf{1}_{[\tau-4]} + \pi_c\,\mathrm{T2Reform}_t \\ + \theta_{c,\tau-5}\,\mathbf{1}_{[\tau-5]} \times \mathrm{T2Reform}_t + \theta_{c,\tau-4}\,\mathbf{1}_{[\tau-4]} \times \mathrm{T2Reform}_t + \varepsilon_{c,t}. \end{aligned} \tag{9}$$

The treated days are $\tau{-}5$ and $\tau{-}4$, the omitted reference is a mid-month control window, and $\mathrm{T2Reform}_t$ equals one from October 6, 2014 onward. The country-level boundary contrast is $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$, and the mechanism predicts $\kappa_c > 0$.

Across the fifteen-market synchronized European sample, the inverse-variance-weighted pooled contrast is +11.85 basis points per day ($t = 2.04$, $p = 0.042$). Twelve of fifteen country estimates have the predicted sign ($p = 0.018$). Because Switzerland transitioned on the same date but is outside the EU and outside the CSDR scope, we also report the EU-only estimate. Excluding Switzerland gives a pooled contrast of +13.71 basis points per day ($t = 2.31$, $p = 0.021$) over fourteen markets.[27]

Internet Appendix Figure IA.10 displays the country-level contrasts as a forest plot. The pooled estimate excludes zero while every individual country's confidence interval is wide; the identification comes from the agreement of the sign across independent markets rather than from any single country. As a falsification test, we estimate the same difference-in-differences at $\tau-4$ versus $\tau-3$, the pair one would choose by mechanically anchoring on the unconditional window's right edge rather than on settlement arithmetic. The estimate is -4.59 basis points per day ($p = 0.43$, untabulated), with only six of fifteen countries positive. The reform therefore reallocates selling pressure only between the days implied by the T+3-to-T+2 settlement transition.

The European evidence complements the U.S. T+1 reform. The U.S. 2024 reform shows that T+2-to-T+1 moves loser-selling pressure from $\tau-4$ to $\tau-3$, while the European 2014 reform shows that T+3-to-T+2 moves it from $\tau-5$ to $\tau-4$. Both reforms move selling pressure one trading day closer to month-end.

5.3 U.S. T+3-to-T+2 Transition

The earlier U.S. transition from T+3 to T+2, effective September 5, 2017, provides an additional test of the same $\tau-5$-to-$\tau-4$ prediction. We estimate the U.S. analogue of the European specification, using $\tau-5$ and $\tau-4$ as the treated boundary days and the early-month control window as the omitted reference.

The estimate is right-signed but statistically insignificant: $\theta_{\tau-5} - \theta_{\tau-4} = +13.08$ basis points ($F = 0.27$, $p = 0.60$; Internet Appendix Table IA.19). As a single-market event, this test is underpowered, and we do not rely on it.[28] The two well-identified reforms,

[27] Country-level estimates, the random-effects (DerSimonian-Laird) variant, date-clustered panel stress tests, false-date placebo distributions, false-boundary placebo tests, and leave-one-country-out estimates are reported in Internet Appendix Section IA.16.

[28] The 2017 transition was widely anticipated and implemented in stages: institutions may have adjusted liquidity buffers and month-end trading routines around the transition window, attenuating a single-date event-study design. Aggregate daily mutual fund flows nevertheless show the predicted one-day shift at this transition, with $\tau-4$ flipping from a net-inflow day under T+3 to a net-outflow day under T+2 (Internet Appendix Table IA.21). We also do not attempt formal identification at the earlier T+5$\rightarrow$T+3 reform in 1995 because institutional ownership and mutual-fund flow pressure were substantially smaller, and because

U.S. 2024 and the synchronized European 2014 transition, carry the settlement evidence that shortening settlement by one day moves loser-selling pressure one trading day closer to month-end.

6 Evidence Beyond the Monthly Cycle

6.1 Liquidity Shocks and the Selection of Loser Stocks

Momentum returns are high during PreTOM because investors engage in dash-for-cash to meet their end-of-month liquidity needs. If institutions' liquidity needs cause selling of loser stocks at the turn-of-the-month, the same asymmetry should appear whenever institutions need cash. That is, losers should fall harder than winners during exogenous liquidity shocks regardless of calendar timing. The fund-level pattern follows mechanically: loser-tilted funds hold what the market sells.

We test this using three institutional liquidity crisis episodes with acute aggregate liquidity needs between 1998 and 2025: September and October of the 2008 Global Financial Crisis, and March 2020 of the COVID-19 pandemic. Table 8, Panel A, documents the effect on stock-level momentum returns. During each of the three crisis episodes, momentum losers dropped heavily, leading to high returns on the WML portfolio. This complements the evidence on causality provided in the previous section.

We next look at mutual funds. At the end of each month, we sort actively managed U.S. equity funds into deciles on trailing 12-month return (months $[m-12, m-1]$, no skip); the decile assignment is predetermined and applies to month m's return. The sort is effectively by funds' momentum loading, since D10 funds hold winners and D1 funds hold losers (Carhart, 1997). We refer to funds in the bottom decile (D1) as "loser funds" and those in the top decile (D10) as "winner funds"; quantities reported at the fund level throughout Sections 6.1 and 7 are aggregated as value-weighted portfolio returns unless noted otherwise. Table 8, Panel B, reports value-weighted monthly excess returns alongside aggregate outflows, and reports each portfolio's *abnormal* return relative to a Carhart (1997) four-factor benchmark (MKT, SMB, HML, UMD), with factor loadings estimated on the 322 non-crisis months in the 1998–2025 sample. The abnormal-return columns isolate the component of each

implementation and data comparability issues make the event less useful as a causal test.

portfolio's crisis-month return that is not explained by linear factor exposure, including momentum (UMD). In March 2020, loser funds underperformed their factor-implied return by 11.17 percentage points while winner funds were essentially at their factor benchmark (-0.26 pp): a 10.92-percentage-point abnormal spread, associated with industry outflows of 1.49% of assets. The 2008 episodes show smaller abnormal spreads (–1.23 pp in September; –2.54 pp in October) but D1 funds' own abnormal returns remain economically large (-3.22 and -3.90 pp). The pattern is consistent with loser-tilted funds bearing the hit because they hold the stocks the market is selling, with the asymmetry strongest in the largest outflow shock.

6.2 International Evidence

We test whether the PreTOM loser effect extends beyond U.S. equities using daily momentum decile portfolios for 20 developed markets from Compustat Global, 1990–2025. Nineteen of these are conventional momentum markets in which past winners outperform past losers on average. The twentieth, Japan, is well-documented to exhibit reversed-sign momentum and is reported separately as a reference rather than pooled with the rest.[29] Within each country-month, we rank stocks into momentum deciles on cumulative returns over months -12 through -2, value-weight by lagged market cap, and subtract the regional value-weighted market return. Regional market returns are taken from Kenneth French's regional factor files for Europe, Asia-Pacific ex-Japan, and Japan.

We pool country-day observations and regress the loser-decile market-adjusted return on a PreTOM indicator with country fixed effects:

$$r_{c,t} - r^m_{c,t} = \alpha_c + \beta \,\mathrm{PreTOM}_t + \varepsilon_{c,t}, \tag{10}$$

where $r_{c,t} - r^m_{c,t}$ is the day-t value-weighted loser-decile return in country c in excess of the regional market return $r^m_{c,t}$, PreTOM_t is the indicator for trading days $[\tau-9, \tau-4]$ (Rest days

[29]Countries, sample periods, and methodology details (size-conditional breakpoints, regional pooling for small markets) are described in Internet Appendix IA.9. Canada is omitted because Compustat Global coverage is limited; Japan's reversed-sign momentum is documented in Fama and French (2012) and Asness et al. (2013). Chui et al. (2000) provide within-Japan evidence that the momentum effect is materially weaker for keiretsu-affiliated firms (which face heavy cross-shareholding and reduced free-float exposure to monthly liquidity-driven trading) than for independent firms, consistent with the broader proposition that institutional ownership structure mediates calendar-time return patterns.

are the omitted reference), α_c are country fixed effects, and standard errors are clustered by country-month. We apply the same $[\tau-9, \tau-4]$ window across all countries rather than tailoring the boundary to each country's settlement regime, which is conservative: local settlement cycles and institutional practices vary, but pooling under a single window avoids country-specific window-tuning and biases against finding an effect to the extent that the binding boundary lies elsewhere in some markets. Table 9 reports the estimates. The coefficient β measures the PreTOM-minus-Rest gap in loser excess returns within country.

In the 19-country pooled sample, losers underperform the regional market by 4.48 bps per day more during PreTOM than during the rest of the month ($t = -4.46$, with standard errors clustered by country-month). The asymmetry is loser-driven: in a similar regression winners show no differential PreTOM concentration (+0.52 bps, $t = 0.54$); the WML spread differs by +5.16 bps ($t = 4.11$). The difference in loser underperformance between PreTOM and Rest is largest in Ireland (−12.32 bps/day), Norway (−11.37), and the Netherlands (−10.57). The effect is present in most of the 19 sample countries but varies in magnitude. Japan, reported separately as a reversed-sign reference, displays the opposite pattern: losers earn slightly higher returns during PreTOM and winners earn lower returns, consistent with the literature documenting reversed-sign momentum in Japan. Country-by-country results are in Internet Appendix IA.9.

7 Revisiting Carhart: Fund Persistence as a PreTOM Phenomenon

Carhart (1997) traces persistent loser-fund underperformance to two components: negative exposure to the momentum factor, UMD, and high expense ratios. Together, these two components account for almost all of what earlier work had attributed to managerial skill.

Our mechanism yields a specific calendar prediction for the first component. Section 3 shows that the WML premium is loser-driven: loser stocks underperform mainly during PreTOM, while winners show no comparable concentration. If the Fama–French UMD factor inherits this same loser-driven calendar structure, then a fund with negative UMD exposure should realize its momentum-loading drag primarily during the same six-day window. We first show that this is the case for UMD itself, and then turn to loser-fund returns. Because the drag comes from exposure to loser stocks, the loser-stock return is the natural control in the fund regressions. It should absorb the PreTOM-concentrated component of D1 fund alpha, while leaving the expense-related drag, which accrues more evenly through the month, largely intact outside PreTOM.

Four implications follow. First, UMD itself should earn most of its return during PreTOM. Second, the momentum-driven component of loser-fund underperformance should concentrate in PreTOM. Third, it should be absorbed by controlling for loser-stock returns. Fourth, the calendar pattern should shift by one day following the T+1 settlement reform. The tests below support these four predictions.

7.1 UMD's Intramonth Profile

We begin by showing the intramonth returns of the UMD factor. The Fama–French UMD factor earns +5.51 bps per day during PreTOM ($t = 3.58$) and only +1.36 bps per day during the remaining fifteen trading days ($t = 1.24$). Over 1980–2025, \$1 invested in UMD only on PreTOM days grows to \$5.51 while \$1 invested only on Rest days grows to \$2.26 (Table 10, Panel A). Any fund with negative UMD exposure realizes the corresponding momentum-loading drag primarily during PreTOM.

7.2 Intramonth Decomposition of D1 Fund Underperformance

We estimate the Carhart regression at daily frequency, splitting the sample into PreTOM and Rest days.[30] For each fund-performance decile Dk, we estimate

$$\begin{aligned} r_d^{Dk} - r_d^f = {} & \alpha_{\text{PreTOM}}^{Dk} \mathbf{1}_{d \in \text{PreTOM}} + \alpha_{\text{Rest}}^{Dk} \mathbf{1}_{d \in \text{Rest}} \\ & + \beta^{Dk} \text{MKT-rf}_d + s^{Dk} \text{SMB}_d + h^{Dk} \text{HML}_d + \gamma^{Dk} r_d^{\text{Loser}} + \varepsilon_d, \end{aligned} \tag{11}$$

where r_d^{Dk} is the equal-weighted day-d net return (after fees and expenses) of funds in performance decile Dk, computed from daily NAV changes plus distributions in the CRSP Mutual Fund Database; r_d^f is the daily one-month T-bill rate; and r_d^{Loser} is the daily value-weighted

[30] We first replicate and extend Carhart (1997)'s loser-fund underperformance result through 2025 in Internet Appendix Section IA.14. Over Carhart's original 1963–1993 sample our estimates mirror his findings. In the extended sample the D10−D1 spread attenuates over time as documented by Choi and Zhao (2021). Daily CRSP mutual fund returns (NAV-implied) are available from September 1998 onward.

return on the bottom stock-momentum decile (D1) from the CRSP stock panel. The first specification sets $\gamma^{Dk} = 0$, giving the standard three-factor alpha. The second allows γ^{Dk} to differ from zero, asking whether exposure to loser stocks absorbs the fund alpha.

We find that in PreTOM, D1 funds earn a three-factor alpha of -2.02 bps per day ($t = -1.57$); after adding the loser-stock return, this falls to -0.38 bps ($t = -0.36$). Outside PreTOM, the corresponding alpha falls only from -2.40 bps ($t = -2.65$) to -1.82 bps ($t = -2.47$). Thus, loser-stock exposure absorbs the PreTOM component of D1 fund underperformance but leaves substantial Rest-day underperformance, consistent with expense drag.

The same mechanism appears on the other side of the fund sort. D1 funds load positively on the loser-stock portfolio ($\beta_{\text{Loser}} = +0.28$), while D10 funds load negatively ($\beta_{\text{Loser}} = -0.19$). D10's positive PreTOM alpha disappears once we control for loser-stock returns. The D1$-$D10 spread therefore reflects a single channel: loser-stock returns hurt funds that hold losers and benefit funds that avoid them (Table 10, Panel B).

7.3 Expense Ratios and the Fund-Level Decomposition

To incorporate fund-specific expense ratios, we move from the portfolio-level time-series regressions in Table 10, Panel B, to a fund-day panel in which each fund remains a separate observation. Table 10, Panel D, decomposes D1 fund underperformance into loser-stock exposure and expense-ratio drag. After both controls are included, D1 funds have no statistically detectable alpha in either window: $+1.85$ bps per day during PreTOM ($t = 1.23$) and -1.10 bps per day during Rest ($t = -1.11$). The loser-stock return absorbs the PreTOM-concentrated component, while expense ratios help account for the remaining underperformance. This is consistent with Carhart's decomposition: the momentum-related component is tied to loser-stock exposure and therefore inherits the PreTOM calendar pattern, whereas expense drag is not itself a calendar-timed trading effect. Thus, the calendar concentration in loser-fund underperformance is the pass-through of the stock-level PreTOM loser cycle to funds that hold those stocks.

7.4 T+1 Evidence on UMD and Loser Funds

The T+1 settlement reform extends the causal evidence to both UMD and the fund level. We re-estimate equation (7) with the daily UMD factor and the equal-weighted D1 fund portfolio replacing $R_t^{D1} - r_t^m$ on the left-hand side, using the same $[\tau{-}4, \tau{-}3]$ treated days and early-month $[\tau{+}5, \tau{+}8]$ control. After the May 2024 reform, UMD's PreTOM concentration shifts from $\tau{-}4$ to $\tau{-}3$. The boundary-day DiD on the daily UMD factor is $\theta_{\tau-3} - \theta_{\tau-4} = +78.1$ bps ($t = 2.48$, $p = 0.013$; Internet Appendix Table IA.12, Panel A).

D1 fund underperformance shows the same one-day shift. The boundary-day DiD on D1 fund excess returns is +29.9 bps ($t = 2.28$) with FF3 controls and +51.6 bps ($t = 2.03$) without controls (Internet Appendix Table IA.12, Panel B). The same regulatory change that moves stock-level loser underperformance one day closer to month-end also moves UMD and loser-fund underperformance by one day. Carhart (1997) shows that loser-fund persistence reflects negative momentum exposure and high expenses, and our evidence shows when the momentum-related component is realized, namely in the PreTOM window.

8 Crash Risk and the PreTOM Premium

A natural concern is that if momentum crashes commonly happen during PreTOM, the PreTOM premium could in part be compensation for crash risk. In contrast, we find that crashes take place predominantly outside the PreTOM period, at month-start.

A trading day is classified as a "crash day" if VW WML falls below −200 basis points. We compare crash incidence against calendar share within a comparison universe that excludes the other window (PreTOM against non-month-start days, month-start against non-PreTOM days), so the two effects do not contaminate each other.[31] Within the non-month-start comparison universe, PreTOM accounts for 30.8% of crash days against its 33.3% calendar share ($z = -1.32$, $p = 0.19$), so the PreTOM premium is not generated by crash avoidance.

Month-start is where crashes concentrate. Month-start days hold a disproportionate share of crash days at every threshold from −100 to −300 bps (Figure 5). The pattern is not driven by well-known momentum-crash episodes: Excluding the 2001 crash (Daniel

[31] Calendar-share construction and the proportions z-test are described in Internet Appendix IA.8.

and Moskowitz, 2016), the March–May 2009 crash, and the COVID-19 crash (February–May 2020) leaves the month-start share at 26.8%, against a 20% calendar share within the non-PreTOM comparison universe, across $N = 522$ remaining crash days ($z = 3.90$, $p < 0.001$; untabulated).[32] Tail-trimming preserves PreTOM profits but flattens month-start returns. This pattern supports the interpretation that PreTOM gains are persistent, while month-start crashes are tail-driven (Internet Appendix Figure IA.4). Figure 6 displays both dimensions jointly with crash probability peaking at month-start and mean WML peaking in PreTOM.

9 Conclusion

The momentum premium is mostly a six-day phenomenon. 77% of WML's cumulative log return is earned during the six trading days before month-end (PreTOM), which comprise just 29% of the trading month. A dollar invested in WML only on PreTOM days grew to \$18.78 between 1980 and 2025; the same dollar invested on the remaining fifteen days grew to \$2.37; over the full sample WML grew to \$44.46. The premium is driven by losers; winners show no comparable concentration.

The mechanism has two pieces. The timing comes from month-end cash demand: predictable payment obligations force investors to raise cash before settlement clears, concentrating selling pressure in the days leading up to the deadline. The cross-sectional incidence falls on losers because they are dispensable on multiple margins: non-dividend-paying, tax-favored for liquidation, and salient under institutional risk reduction. Three pieces of evidence connect mechanism to result: TAQ trade-direction data show institutional selling pressure on losers concentrated in PreTOM; the May 2024 T+1 settlement reform shifted the selling window by one day, with placebo tests returning null; and the loser-asymmetric pattern replicates across 19 developed markets and in three episodes of acute fund outflows (September 2008, October 2008, March 2020).

The findings reframe the bulk of momentum returns as a feature of market financial plumbing rather than a property of investor beliefs or risk. The same logic may generalize: many cross-sectional anomalies share a short leg populated by stocks that investors would

[32]Methodology and list of excluded months are detailed in Internet Appendix Section IA.8; the post-exclusion numbers themselves are not tabulated.

sell first under liquidity pressure, and if the intramonth structure documented here extends to those anomalies, a portion of cross-sectional return predictability may turn out to be a settlement-cycle phenomenon.

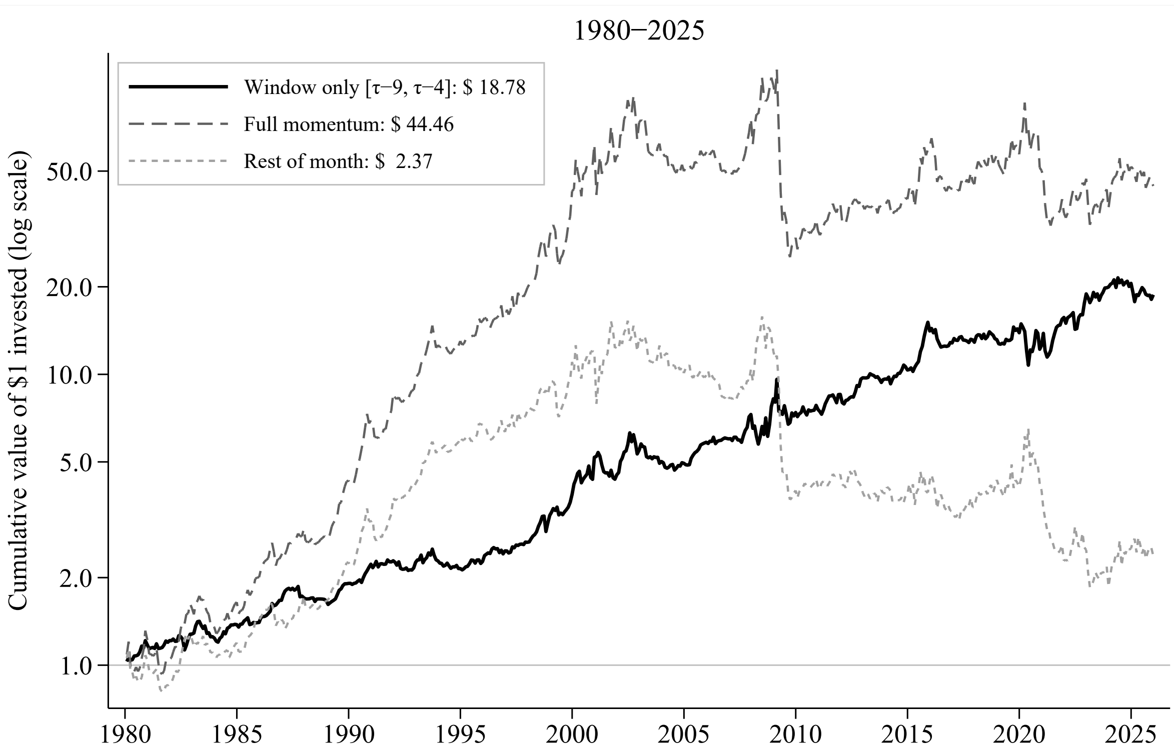

Figure 1: Cumulative wealth from $1 invested in three momentum strategies: WML earned only during PreTOM ($[\tau-9, \tau-4]$), WML earned only during Rest (i.e., the approximately 15 trading days outside the PreTOM window), and WML earned every day. WML is a self-financing long-short portfolio; on days outside the relevant window the strategy holds cash at zero return. Trading days are indexed relative to the turn-of-the-month (month-end) denoted by τ. Daily returns are value-weighted, 1980–2025. Deciles are formed each month by ranking all NYSE, AMEX, and NASDAQ stocks in CRSP on cumulative returns over months -12 through -2 using NYSE breakpoints, and held constant within the month.

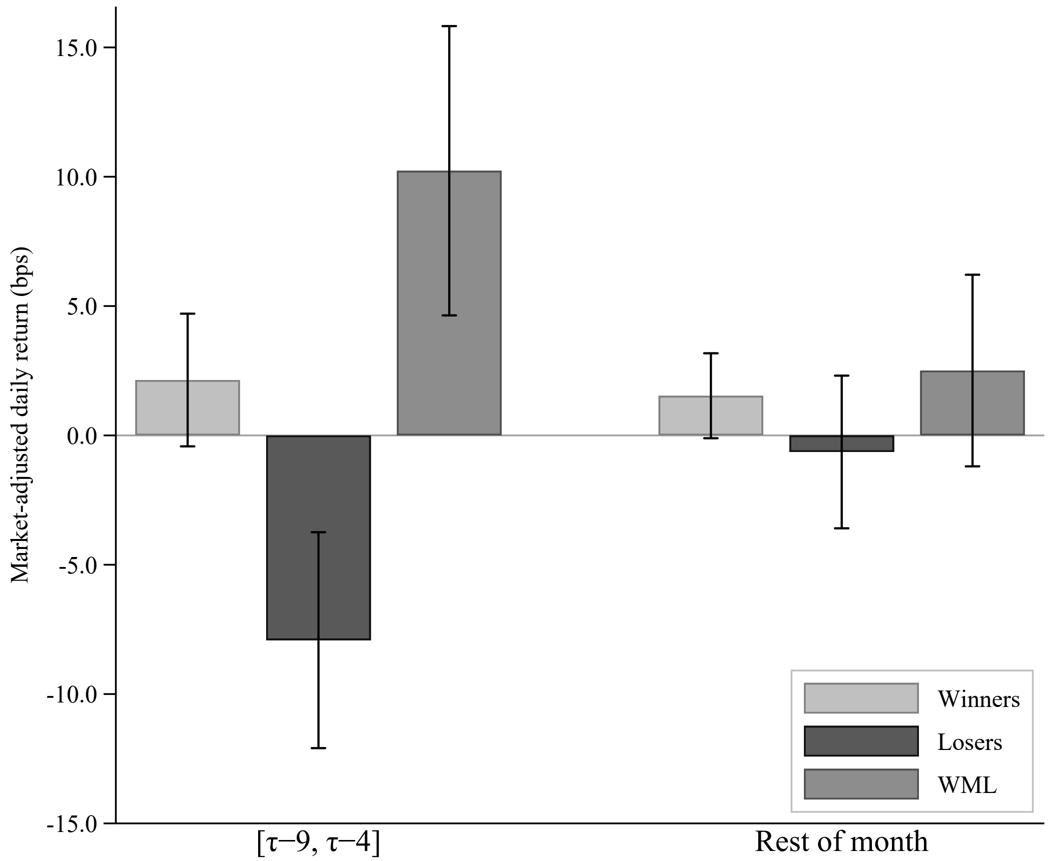

Figure 2: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during PreTOM ($[\tau{-}9, \tau{-}4]$) versus Rest. Daily returns are compounded within each window and averaged across months; 95% confidence intervals use heteroskedasticity-robust standard errors. Daily returns are value-weighted. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025.

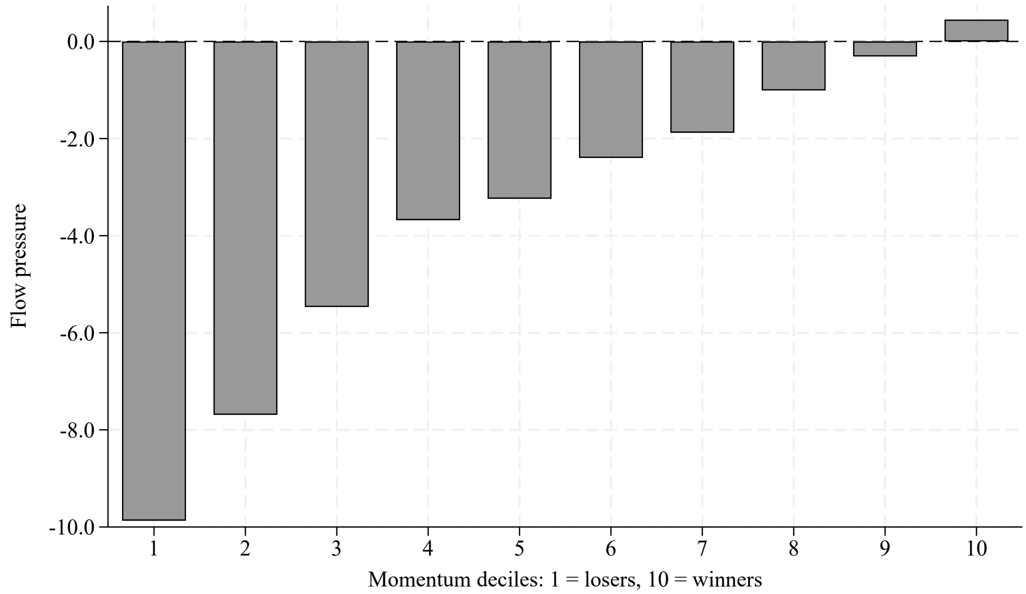

Figure 3: Cumulative mutual fund flow pressure during PreTOM ($[\tau - 9, \tau - 4]$) by momentum decile. Flow pressure is calculated from daily mutual fund flows allocated to stocks in proportion to prior-quarter holdings, normalized by market capitalization. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 2010–2023.

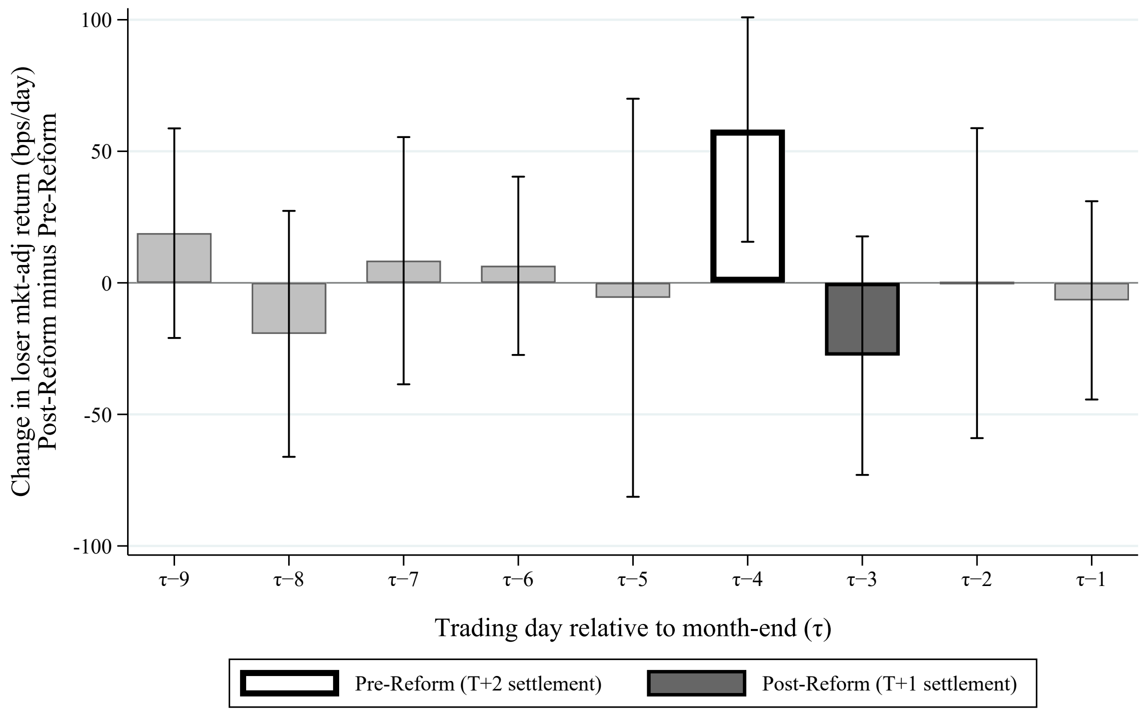

Figure 4: Difference in mean daily market-adjusted VW loser returns (Post-T+1 minus Pre-T+1) at each trading day relative to month-end ($\tau-9$ through $\tau-1$), with 95% confidence intervals. The SEC transitioned from T+2 to T+1 settlement on May 28, 2024. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025 (532 months pre, 19 months post).

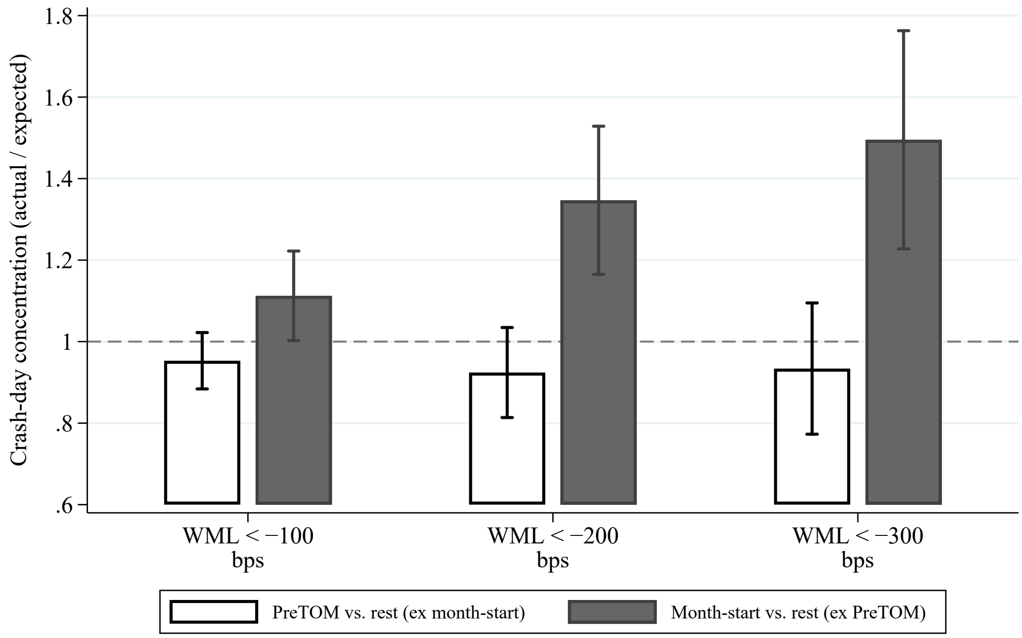

Figure 5: Crash-day concentration by calendar window. Each point plots the ratio of actual crash-day share to calendar share; whiskers show 95% confidence intervals. The dashed horizontal line marks a ratio of one, the value expected if crash days were distributed uniformly across the calendar. Daily WML returns are value-weighted. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025.

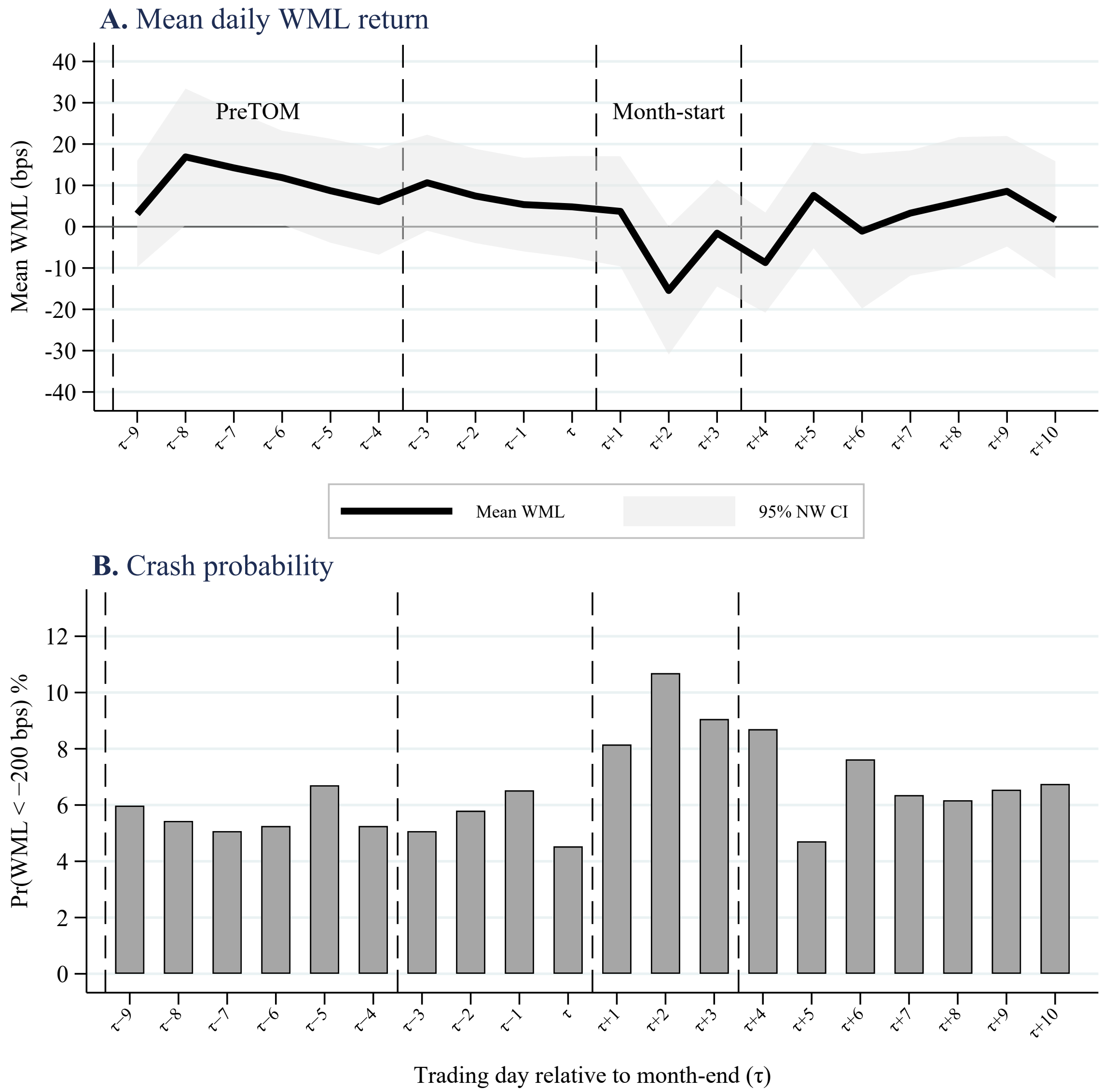

Figure 6: Panel A plots mean daily WML return with Newey-West 95% confidence bands by trading day $\tau+i$ relative to the turn-of-the-month τ. Panel B plots crash probability, defined as Pr(WML $<$ -200 bps). Dashed vertical lines mark the PreTOM window ($[\tau-9, \tau-4]$) and the month-start window ($\tau+1$ to $\tau+3$). Daily returns are value-weighted. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025.

Table 1: Calendar-Conditional Momentum-Decile Premia: Fama–MacBeth Estimates

	(1) Losers (D1)	(2) Winners (D10)
Rest average (c_0^d)	-0.866	1.522^{*}
	(-0.58)	(1.78)
PreTOM differential (c_1^d)	-7.116^{***}	0.646
	(-2.79)	(0.39)
PreTOM average ($c_0^d + c_1^d$)	-7.982^{***}	2.168
	(-3.75)	(1.63)
Sample	1980–2025	1980–2025
Observations	11,595	11,595

Notes. Daily Fama–MacBeth estimates from equations (1)–(2). Stage 1 regresses stock-level market-adjusted returns on momentum-decile indicators with no intercept, value-weighted by lagged market capitalization, across all CRSP common stocks each trading day. Stage 2 regresses the daily decile coefficient on a PreTOM indicator with Newey–West standard errors at 21 lags. PreTOM is the trading-day window $[\tau-9, \tau-4]$. Sample: 1980–2025. Coefficients are given in basis points; t-statistics are in parentheses; standard errors are Newey–West. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Loser PreTOM Underperformance: Baseline Panel Regression

	(1) VW	(2) EW
Loser (β_1)	2.609**	8.022***
	(2.04)	(11.19)
Loser $\times$ PreTOM (β_2)	−7.151***	−2.550*
	(-3.08)	(-1.85)
Sample	1980–2025	1980–2025
Fixed effects	Stock, Date	Stock, Date
Observations	53,324,145	53,324,145

Notes. Estimates of equation (3) on the full CRSP panel of NYSE/AMEX/NASDAQ stocks, 1980–2025. Dependent variable is the stock's daily excess return over the value-weighted CRSP market, $\text{ExRet}_{i,t} = r_{i,t} - r_t^m$, in basis points. $\text{Loser}_{i,t}$ is an indicator equal to one if stock i is in the bottom momentum decile on day t. PreTOM_t is an indicator for the six trading days $[\tau-9, \tau-4]$ before month-end (τ is the last trading day). All specifications include stock and date fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. Column 1 weights observations by lagged market capitalization within decile-date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: PreTOM Effect on Losers and the Bid-Ask Spread

	(1) VW	(2) EW
PreTOM (β)	-9.039^{***}	-4.237^{*}
	(-2.92)	(-1.96)
$\text{BAS}_{\text{prior}}$ (γ_1)	99.510^{***}	247.814^{***}
	(8.08)	(24.68)
PreTOM $\times$ $\text{BAS}_{\text{prior}}$ (γ_2)	39.040	29.537^{***}
	(1.41)	(2.82)
Sample	1980–2025	1980–2025
Fixed effects	Stock, Month	Stock, Month
Observations	9,069,541	9,069,541

Notes. Sample: bottom-decile (D1) momentum stocks, 1980–2025. Dependent variable is $\text{ExRet}_{i,t} = r_{i,t} - r_t^m$ (market-adjusted return) in basis points, as defined in equation (3). PreTOM is an indicator for the six trading days $[\tau-9, \tau-4]$ before month-end (τ is the last trading day). $\text{BAS}_{\text{prior}}$ is the mean bid-ask spread of stock i over calendar month $m-1$, as a fraction of the quote midpoint, computed from all CRSP stocks regardless of decile membership in month $m-1$; stock-months with fewer than 10 valid prior-month BAS days are excluded. The estimation sample is the full D1 panel with a valid $\text{BAS}_{\text{prior}}$; no restriction is placed on the availability of the current-day spread. All specifications include stock and year-month fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. Column 1 weights observations by lagged market capitalization. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Subperiod Stability of the PreTOM Effect

	(1) 1980–2002	(2) 2002–2025
Loser	0.345	6.063***
	(0.27)	(2.91)
Loser × PreTOM	−5.762**	−8.503**
	(-2.31)	(-2.20)
Fixed effects	Stock, Date	Stock, Date
Observations	30,596,232	22,727,911
Within R^2	0.0000	0.0001

Notes. Estimates of equation (3) on subperiods. Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^m$, in excess of the value-weighted CRSP market return), in basis points. Returns are value-weighted. Sample split at midpoint (July 2002). All specifications include stock and date fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Selling Pressure During PreTOM

	(1)
Loser (β_1)	0.0122***
	(20.20)
Loser $\times$ PreTOM (β_2)	0.0020**
	(2.50)
Sample	2003–2022
Fixed effects	Stock, Date
Observations	17,861,276

Notes. Estimates of equation (5) on the full panel of CRSP/TAQ stocks, 2003–2022. The dependent variable is net selling pressure = (sell volume − buy volume)/total volume, which by construction ranges from −1 to +1. Trade direction is assigned using the Lee–Ready algorithm (WRDS Intraday Indicators). $\text{Loser}_{i,t}$ is an indicator for the bottom momentum decile. PreTOM_t is an indicator for $[\tau-9, \tau-4]$ before month-end (τ is the last trading day). The specification includes stock and date fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Partial Reversal of PreTOM Loser Underperformance

	(1) Full Sample
Loser	1.043
	(0.55)
Loser × PreTOM	-5.585^{**}
	(-2.07)
Loser × Post	3.356
	(1.24)
Fixed effects	Stock, Date
Observations	53,324,145
Within R^2	0.0000

Notes. Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^m$, in excess of the value-weighted CRSP market return), in basis points. Loser equals one for bottom-decile momentum stocks (fixed monthly sorting). PreTOM equals one during trading days $\tau-9$ to $\tau-4$ relative to month-end (τ is the last trading day). Post equals one during trading days $\tau-3$ to $\tau+3$ (month-end). Returns are value-weighted (lagged market cap). All specifications include stock and date fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Settlement Change Difference-in-Differences

Panel A: Cell Means (loser-decile market-adjusted return, bps per day)	Pre T+1 settlement / Post T+1 settlement
Early-month $[\tau{+}5, \tau{+}8]$ (control)	-3.51 / $+7.04$
$\tau{-}3$	-3.69 / -31.36
$\tau{-}4$	-6.90 / $+51.36$

Panel B: Portfolio-Level DiD (early-month control)	(1) Pre-period: T+2 settlement era only (Sep 2017–Apr 2024)	(2) Pre-period: full history (1980–Apr 2024)
$\beta_{\tau-4}$: $\mathbf{1}_{[\tau-4]}$	-22.25	-3.39
	(16.82)	(4.82)
$\beta_{\tau-3}$: $\mathbf{1}_{[\tau-3]}$	-19.30	-0.18
	(14.36)	(5.49)
π: T1Reform_t	7.78	10.55
	(21.31)	(18.67)
$\theta_{\tau-4}$: $\mathbf{1}_{[\tau-4]} \times \text{T1Reform}_t$	66.56^{*}	47.71
	(38.54)	(34.53)
$\theta_{\tau-3}$: $\mathbf{1}_{[\tau-3]} \times \text{T1Reform}_t$	-19.10	-38.22
	(31.83)	(28.66)
$\theta_{\tau-4} - \theta_{\tau-3}$ (one-day shift)	85.66^{**}	85.93^{***}
	(37.07)	(32.01)
Observations	297	1,653

Notes. Estimates of equation (7). Dependent variable is the daily market-adjusted value-weighted loser portfolio return in basis points. Treated days are $\tau{-}4$ and $\tau{-}3$ (τ is the last trading day of the month); control days are early-month $[\tau{+}5, \tau{+}8]$, averaged to one observation per month, so each month contributes three observations: $\tau{-}4$, $\tau{-}3$, and one early-month-control mean. T1Reform_t equals one for dates from June 1, 2024 onward, after the T+1 settlement reform took effect; May 2024 is excluded from both pre- and post-samples to avoid contamination at the regime boundary. Column 1 uses the T+2 settlement era (Sep 2017 to Apr 2024) as the pre-period; Column 2 uses 1980 to Apr 2024. Panel A reports cell means. Panel B reports portfolio-level OLS coefficients with Newey-West standard errors in parentheses (NW(10) lags). Newey-West t-statistics on the differential are stable across NW(5), NW(10), and NW(21) lag choices. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Settlement Change Difference-in-Differences (continued)

Panel C: Stock-Level Triple-Difference (D1 and D10, stock and day fixed effects, early-month control)	(1) Pre-period: 1980 to Apr 2024
$\theta_{\tau-4}$: $\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-4]} \times \text{T1Reform}_t$	161.24**
	(73.23)
$\theta_{\tau-3}$: $\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-3]} \times \text{T1Reform}_t$	−35.07
	(52.70)
$\theta_{\tau-4} - \theta_{\tau-3}$ (one-day shift)	196.32**
	(89.32)
Observations	4,907,237

Notes. Stock-level triple-difference of equation (8). Sample: D1 (loser) and D10 (winner) decile stocks; pre-period 1980 to Apr 2024, post-period from June 1, 2024 (May 2024 excluded). Dependent variable is the daily market-adjusted return in basis points. Returns are value-weighted by lagged market capitalization. Standard errors two-way clustered by stock and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. See Table 7 for details. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Liquidity Shocks and Asymmetric Loser Selling

Panel A: Stock-Level Momentum Portfolio Returns During Liquidity Crisis Episodes

Episode	(1) D1 (Losers) (%)	(2) D10 (Winners) (%)	(3) D10−D1 (pp)
Sep 2008 (GFC)	−25.90	−15.42	+10.48
Oct 2008 (GFC)	−26.05	−15.12	+10.93
Mar 2020 (COVID)	−27.96	−9.77	+18.19

Panel B: Momentum Fund Returns During the Same Three Episodes

Episode	(1) D1 excess (%)	(2) D1 Carhart abnormal (%)	(3) D10 excess (%)	(4) D10 Carhart abnormal (%)	(5) D1−D10 Carhart abnormal (pp)
Sep 2008 (GFC)	−11.56	−3.22	−12.52	−1.99	−1.23
Oct 2008 (GFC)	−21.89	−3.90	−16.01	−1.36	−2.54
Mar 2020 (COVID)	−24.92	−11.17	−11.29	−0.26	−10.92

Notes. Panel A reports compounded value-weighted monthly returns on stock-level momentum decile portfolios (NYSE breakpoints, fixed monthly sorts on cumulative returns over months −12 through −2) during three acute-outflow episodes; the D10−D1 column is the percentage-point spread between winner and loser deciles. Panel B reports value-weighted (lagged TNA) monthly returns on D1 and D10 mutual fund portfolios during the same three episodes. Aggregate mutual fund net outflows during these episodes were 0.90% (September 2008), 1.79% (October 2008), and 1.49% (March 2020) of one-month-lagged assets. Funds are sorted at the end of each month into deciles on their trailing 12-month return (months $[m-12, m-1]$, no skip); the decile assignment applies to month m's return. The "excess" columns report monthly returns in excess of the one-month T-bill rate; the "Carhart abnormal" columns report the residual after subtracting the return implied by a Carhart (1997) four-factor model (MKT, SMB, HML, UMD), with loadings estimated on the 322 non-crisis months in the 1998–2025 sample. The D1−D10 Carhart abnormal column is the difference of the two abnormal-return columns. Sample for Panel B: CRSP Mutual Fund Database, actively managed U.S. equity diversified funds (objective code ED, AUM $\geq$ \$10M, excluding index funds, ETFs, and ETNs) that appear in either D1 or D10 in at least one month between 1998 and 2025; all months for those funds are retained.

Table 9: International Evidence: PreTOM Returns across Developed Markets

	PreTOM		Rest of Month		Difference	
	Mean	t	Mean	t	Mean	t
Panel A: Pooled regression (19 countries, 1990–2025)						
Losers − Market	−4.76	(−5.82)	−0.28	(−0.49)	−4.48	(−4.46)
Winners − Market	+3.44	(4.35)	+2.92	(6.01)	+0.52	(0.54)
WML	+8.27	(7.77)	+3.10	(4.70)	+5.16	(4.11)
Panel B: Selected countries — Losers − Market (bps/day)						
Norway	−12.10	(−3.26)	−0.73	(−0.32)	−11.37	(−2.63)
Netherlands	−7.93	(−1.96)	+2.65	(1.14)	−10.57	(−2.31)
Sweden	−4.62	(−1.19)	+3.68	(1.71)	−8.30	(−1.90)
Switzerland	−4.59	(−0.98)	+1.86	(0.78)	−6.45	(−1.19)
Belgium	−6.55	(−2.08)	−1.78	(−0.61)	−4.77	(−1.11)
Hong Kong	−9.15	(−2.73)	−0.21	(−0.09)	−8.94	(−2.21)
UK	−3.09	(−1.06)	−1.35	(−0.65)	−1.74	(−0.45)
France	−4.85	(−1.77)	−0.18	(−0.08)	−4.67	(−1.35)
Germany	−5.05	(−1.81)	−2.75	(−1.26)	−2.30	(−0.63)
Australia	−8.23	(−3.28)	−1.56	(−0.84)	−6.67	(−2.24)
Panel C: Reversed-sign reference (excluded from pooled sample)						
Japan: Losers − Market	+0.68	(0.31)	+0.12	(0.08)	+0.56	(0.21)
Japan: Winners − Market	−0.90	(−0.45)	+1.88	(1.42)	−2.78	(−1.12)
Japan: WML	−1.59	(−0.53)	+1.76	(0.97)	−3.35	(−0.97)

Notes: Daily value-weighted portfolio returns from Compustat Global, 1990–2025 (HKG/CHE from 1993). Twenty developed markets excluding Canada; the pooled regression and the per-country panel cover the 19 conventional momentum markets, with Japan reported separately as a reversed-sign reference because momentum signs reverse there (Fama and French, 2012; Asness et al., 2013). Momentum deciles are formed within each country-month using cumulative returns over months −12 through −2. Decile breakpoints are set using stocks above the within-country median market capitalization (analogous to NYSE breakpoints in the U.S. analysis); all stocks are then assigned to deciles based on those cutoffs. For countries with fewer than 50 qualifying stocks, breakpoints are pooled within the broader region. Returns are market-adjusted using Kenneth French's regional factor files. PreTOM denotes trading days $T-9$ through $T-4$ relative to month-end; Rest is all other days; Difference is PreTOM minus Rest. Panel A reports pooled estimates with country fixed effects and standard errors clustered by country-month. Panel B reports per-country estimates for the ten largest momentum markets, with standard errors clustered by month. Panel C reports Japan separately, with the same per-country specification. WML returns are computable on all country-days, while Loser and Winner market-adjusted returns require a non-missing regional market return; the sample for WML is therefore slightly larger, which produces the small non-additivity between Winner − Loser and WML in the reported coefficients.

Table 10: UMD Factor and Carhart Decomposition at Daily Frequency

Panel A: Daily UMD returns by calendar window, 1980–2025

	Mean (bps/day)	t-stat	\$1 cumulative	Days
PreTOM ($[\tau-9, \tau-4]$)	5.51	3.58	\$5.51	3,312
Rest of month	1.36	1.24	\$2.26	8,283
Full month	2.55	2.84	\$12.44	11,595

Panel B: D1 and D10 fund daily alphas (bps/day) by calendar window, with loser-stock excess return control (1998–2025)

	D1 (Losers)		D10 (Winners)	
Specification	α	β_{Loser}	α	β_{Loser}
PreTOM only				
FF3 only	-2.02		1.49	
	(-1.57)		(1.30)	
FF3 + Loser	-0.38	0.28^{***}	0.39	-0.19^{***}
	(-0.36)	(10.88)	(0.37)	(-9.38)
Observations	1,968	1,968	1,968	1,968
Rest of the month				
FF3 only	-2.40^{***}		1.31^{*}	
	(-2.65)		(1.69)	
FF3 + Loser	-1.82^{**}	0.28^{***}	0.93	-0.18^{***}
	(-2.47)	(18.01)	(1.36)	(-13.06)
Observations	4,908	4,908	4,908	4,908
Full sample				
FF3 only	-2.28^{***}		1.32^{**}	
	(-3.10)		(2.06)	
FF3 + Loser	-1.41^{**}	0.28^{***}	0.75	-0.18^{***}
	(-2.32)	(20.36)	(1.32)	(-15.49)
Observations	6,876	6,876	6,876	6,876

Notes. Panel A decomposes the Fama–French daily UMD factor into PreTOM and Rest-of-month components, 1980–2025; t-statistics use Newey–West HAC standard errors with automatic bandwidth. Panel B reports estimates of equation (11) for funds in the loser (D1) or winner (D10) decile, 1998–2025. The LHS is the equal-weighted daily fund excess return (fund return minus the one-month T-bill rate) in basis points; daily fund returns outside ±50% are dropped. The loser-stock control is the value-weighted excess return on the bottom CRSP momentum decile of stocks (also in fund-return form: stock return minus r^f), entered as a level—not residualized against the FF3 factors. HAC(5) Newey–West time-series regressions; β_{Loser} measures the fund portfolio's loading on the loser-stock excess return. t-statistics are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: UMD Factor Decomposition and Carhart Replication (continued)

Panel D: D1 (losers only) fund alpha — stepwise absorption by Loser return and expense ratio

	(1) FF3	(2) +Loser	(3) +Loser+Exp
PreTOM days (N = 1,125,059)			
α (bps/day)	-1.87	-0.19	$+1.85$
	(-1.56)	(-0.19)	$(+1.23)$
β_{Loser}	—	$+0.264^{***}$	$+0.270^{***}$
	—	$(+15.18)$	$(+14.95)$
Exp β (per 1%)	—	—	-1.48^{*}
	—	—	(-1.95)
Rest of the days (N = 2,806,032)			
α (bps/day)	-2.35^{***}	-1.95^{***}	-1.10
	(-2.98)	(-3.00)	(-1.11)
β_{Loser}	—	$+0.259^{***}$	$+0.267^{***}$
	—	$(+26.05)$	$(+25.76)$
Exp β (per 1%)	—	—	-0.59
	—	—	(-1.19)

Notes (Panel D): Fund-day panel within D1, N as indicated; standard errors clustered by date; expense ratio per 1% annual. The underlying fund universe, date range (1998–2025), and return filter ($[-50\%, +50\%]$) are identical to Panel C, but observations here are at the fund-day level rather than the portfolio (day) level, which allows fund-specific covariates such as the expense ratio. t-stats in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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